IDIS

Web API Protocol

Ver. 2.20

2022-11-17

Document History

Version	Date	Comment
2.0	2017-02-01	Added Resolutions and Fixed VideoExposure
2.1	2017-06-29	Added easy video setting
2.2	2017-07-24	Added sensor mode & fish eye (group6)
2.3	2017-08-16	Added video snapshot(group 5,6)
2.4	2017-08-18	Added smart IR
2.5	2017-11-23	Added manualAlarmOut and intelligent codec level / Document error correction.
2.6	2017-11-24	Added audio noise filter.
2.7	2017-12-22	Added PIR detection event / Document error correction
2.8	2017-12-26	Added multiple alarm in/alarm out function
2.9	2017-12-26	Added Auto tracking event
2.10	2017-01-11	Added multiple alarm out settings for event action. & Fixed document error
2.11	2017-04-04	Added a device name constraint rule. / Fixed document error.
2.12	2019-10-02	Added LensReset for MFZ lens.
2.13	2020-03-03	Added wiper on function to PTZ command.
2.14	2020-05-20	Added Get/Set absolute PTZ position / Move to Point
	2020-05-27	Added TCP packet message format.
2.15	2020-08-31	Added Get/Set absolute PTZ position (Extended version) Modified Set Event Nofitication.
2.16	2021-11-10	Added OSD text/datetime, Link-local only, WDR level, Onvif event type.
2.17	2022-03-07	Added PTZ Move to Area (ROI).
2.18	2022-04-13	Modified Time Zone Name.
2.19	2022-04-29	Added IPv6 address, Sensor frame rate, Rtp multicast. Fixed document error.
2.20	2022-11-17	Added Debug Log, Modified System Log

0. Introductions

1. The time represented on the API is based on local machine (except Chapter 2. time parameter)

2. The API message is formed as follows : <parameter>=<value>[&<parameter>=<value>…]

3. Requested parameters which are not supported on target camera will not be returned.

4. Characters in the <value> must be URI encoded, except alphabet and number, "-", "_", ".", "~"

ex> language=english&note=!@#$%^&*()=

→ language=english&note=%21%40%23%24%25%5E%26%2A%28%29%3D

5. The API is based on the standard HTTP Authentication. (refer RFC2617)

6. You can access the camera and monitor live video images using media players, such as VLC Player, supporting RTSP service. You should open all ports for UDP protocol or an RTSP port for TCP protocol if the camera uses a NAT (Network Address Translation) device for network connection or if the firewall is enabled. This function might not be supported, depending on the type of media player, and some media players might not play video properly depending on network conditions or compression or resolution of images for streaming.

- You can access video as follows:
  Access from a PC: Start the media player and enter "*rtsp://*user*:*Password@IP address*:*RTSP port number*/trackID=*'stream number'"
  (the number for the primary stream is 1, for the secondary stream is 2, and for the tertiary stream is 3)

example: *rtsp://admin:@10.0.152.35:554/trackID=1*
(user: *admin*, password: no password, camera IP address: *10.0.152.35*, RTSP port number: *554*, stream: primary)

- You can access audio as follows:
  Access from a PC: Start the media player and enter "*rtsp://*user*:*Password@IP address*:*RTSP port number*/trackID=*'*max* stream number+1'"

example: *rtsp://admin:@10.0.152.35:554/trackID=4*
(user: *admin*, password: no password, camera IP address: *10.0.152.35*, RTSP port number: *554*,
audio: if max video stream is 3, 4 mean audio stream)

1. How to Read

- Action template of send parameters

Parameters	**Description**	**Values**
action	Action ID	<action ID>
<send parameter 1>	Description of the <send parameter 1>	<send value 1> Definition of the <parameter name1>'s valid values and type.
<send parameter 2>	…	<send value 2>
…	…	..
<send parameter N>	…	<send value N>

- Action template of receive parameters

Parameters	**Description**	**Values**
returnCode	Return code	<return code> Refer Return Code
<receive parameter 1>	Description of the <receive parameter 1>	<receive value 1> Definition of the <parameter name1>'s valid values and type.
<receive parameter 2>	…	<receive value 2>
…	…	..
<receive parameter N>	…	<receive value N>

- Example of Action Template

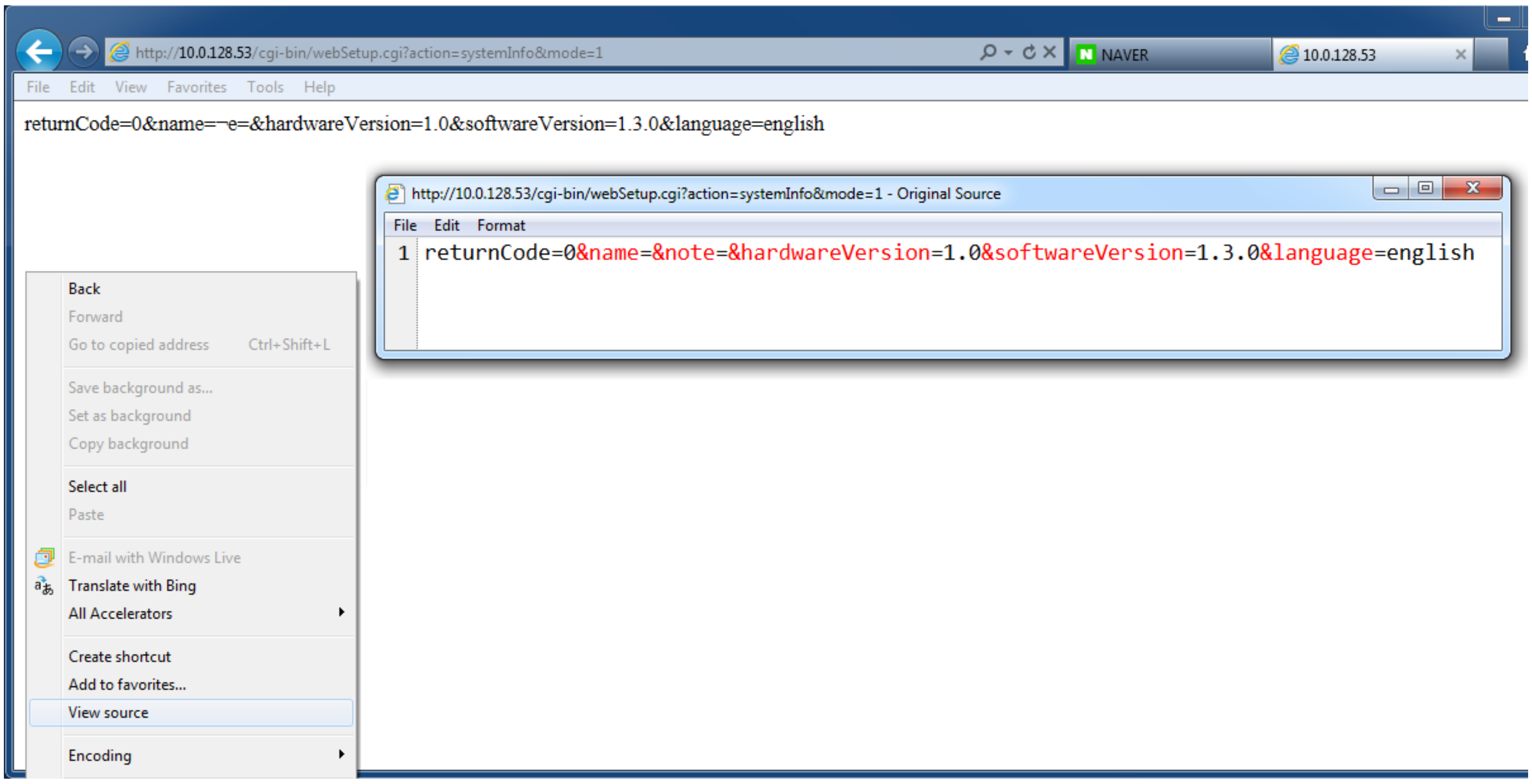
Request:
action=<action ID>&<send parameter 1>=<send value 1>& … &<send parameter N>=<send value N>

Response:
returnCode=<return code>&<receive parameter 1>=<receive value 1>& … &<receive parameter N>=<receive value N>

For example, you can test the API by using web broswer.

Request:
action=systemInfo&mode=1

Response:
returnCode=0&name=&note=&hardwareVersion=1.0&softwareVersion=1.1.7&statusLed=on&language=english



READ example for exposure setup
http://10.0.17.15/cgi-bin/webSetup.cgi?action=videoExposure&mode=1

WRITE example for exposure setup
http://10.0.17.15/cgi-bin/webSetup.cgi?action=videoExposure&mode=0&targetGain=0&localExposure=off&antiFlicker=off&slowShutter=off&manualAeControl=on&lowerShutterLimit=1000&upperShutterLimit=1000&lowerGainLimit=iso400&upperGainLimit=iso400&irisControlMode=fullopen

2. System Information

2.1. Read

Get system information.

Method: GET

Syntax

`http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]`

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	systemInfo	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
name	Device name (Group 1~4 : Only English, numbers and special characters are supported.)	<string> ex> name=NC-B320IP	
note	Device description	<string> ex> note=enterance	
hardwareVersion	SW Version	<string> ex> softwareVersion=1.0	Read only
softwareVersion	HW Version	<string> ex> hardwareVersion=1.0	Read only
useOnvif	support ONVIF on/off	on | off	
statusLed	Status LED on/off	on | off	
onvifEventType	Onvif Event Type	standard | normal	
language	Language	english | french | german | italian | spanish | dutch | polish | portuguese | hungarian | czech | russian | korean | japanese | chinese-PRC | chinese-Taiwan | danish | swedish | finnish | turkish | croatian	
predefined Mode	set predefined mode	~~normal | mode1~~ off | normal_slower | normal_slow | normal_normal | normal_fast | normal_faster | localexposure_auto_slower | localexposure_auto_slow | localexposure_auto_normal | localexposure_auto_fast | localexposure_auto_faster | localexposure_on_slower | localexposure_on_slow | localexposure_on_normal | localexposure_on_fast | localexposure_on_faster | high_iso_slower | high_iso_slow | high_iso_normal | high_iso_fast | high_iso_faster	Predefined mode support model

Example

Send data:
action=systemInfo&mode=1

Response data:
returnCode=0&name=NC-B320IP&note=&hardwareVersion=1.0&softwareVersion=1.1.7&useOnvif=on&onvifEventType=standard&statusLed=on&language=english

2.2. Write

Set system information.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	systemInfo	
mode	Mode Code	0	
name	Device name (Group 1~4 : Only English, numbers and special characters are supported.)	<string> ex> name=NC-B320IP	
note	Device description	<string> ex> note=enterance	
useOnvif	support ONVIF on/off	on | off	
statusLed	Status LED on/off	on | off	
onvifEventType	Onvif Event Type	standard | normal	
language	Language	english | french | german | italian | spanish | dutch | polish | portuguese | hungarian | czech | russian | korean | japanese | chinese-PRC | chinese-Taiwan | danish | swedish | finnish | turkish | croatian	
predefined Mode	set predefined mode	~~normal | mode1~~ off | normal_slower | normal_slow | normal_normal | normal_fast | normal_faster | localexposure_auto_slower | localexposure_auto_slow | localexposure_auto_normal | localexposure_auto_fast | localexposure_auto_faster | localexposure_on_slower | localexposure_on_slow | localexposure_on_normal | localexposure_on_fast | localexposure_on_faster | high_iso_slower | high_iso_slow | high_iso_normal | high_iso_fast | high_iso_faster	Predefined mode support model

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=systemInfo&mode=0&name=sampleCamera&note=enterance&useOnvif=on&statusLed=on&language=english&onvifEventType=standard

Response data:

returnCode=0

3. Date/Time

3.1. Read

Get system date, time and language.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	dateTime	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
dateFormat	Date format	YYYY/MM/DD | MM/DD/YYYY | DD/MM/YYYY ex> MM/DD/YYYY → 11/04/2013	
timeFormat	Time format	HH:MM:SS | II:MM:SS_PP | PP_II:MM:SS ex> HH:MM:SS → 13:11:12 PP_II:MM:SS → PM 01:11:12	
time	System date and time (based on UTC)	<string> YYYY/MM/DD_HH:MM:SS ex> 2013/12/03_19:20:11	
time	System date and time (based on camera time)	<string> YYYY/MM/DD_HH:MM:SS ex> 2013/12/03_19:20:11	Group 5~6
timeZone	Time zone	Refer Time_Zone ex> timeZone=seoul	
dst	Use daylight saving time	on | off ex> dst=off	
timeSync	Use automatic time sync	on | off ex> timeSync=off	
timeServer	NTP server name	<string> ex> timeServer=time.kriss.re.kr	
useDDNS	Use NTP server name from DDNS registered	on | off ex> useDDNS=off	
timeSyncInterval	Time sync interval (minute)	30 | 60 | 120 | 180 | 360 | 540 | 720 | 1440 ex> timeSyncInterval=60	
runAsServer	Run as NTP server	on | off ex> runAsServer=off	

Example

Send data:

action=dateTime&mode=1

Response data:

returnCode=0&dateFormat=YYYY/MM/DD&timeFormat=HH:MM:SS&time=2013/12/03_19:20:11&timeZone=seoul&dst=off&timeSyn c=on&timeserver=time.com&useDDNS=off&timeSyncInterval=60&runAsServer=off

3.2. Write

Set system date, time and language.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	dateTime	
mode	Mode Code	0	
dateFormat	Date format	YYYY/MM/DD | MM/DD/YYYY | DD/MM/YYYY ex> MM/DD/YYYY → 11/04/2013	
timeFormat	Time format	HH:MM:SS | II:MM:SS_PP | PP_II:MM:SS ex> HH:MM:SS → 13:11:12 PP_II:MM:SS → PM 01:11:12	
time	System date and time (based on UTC)	<string> YYYY/MM/DD_HH:MM:SS ex> 2013/12/03_19:20:11	Group 1~4
time	System date and time (based on camera time)	<string> YYYY/MM/DD_HH:MM:SS ex> 2013/12/03_19:20:11	Group 5~6
timeZone	Time zone	Refer Time_Zone ex> timeZone=seoul	
dst	Use daylight saving time	on | off ex> dst=off	
timeSync	Use automatic time sync	on | off ex> timeSync=off	
timeServer	NTP server name	<string> ex> timeServer=time.kriss.re.kr	
useDDNS	Use NTP server name from DDNS registered	on | off ex> useDDNS=off	
timeSyncInterval	Time sync interval (minute)	30 | 60 | 120 | 180 | 360 | 540 | 720 | 1440 ex> timeSyncInterval=60	
runAsServer	Run as NTP server	on | off ex> runAsServer=off	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=dateTime&mode=0&dateFormat=YYYY/MM/DD&timeFormat=HH:MM:SS&time=2013/12/03_19:20:11&timeZone=Seoul&dst=off&timeSync=on&timeServer=time.com&useDDNS=off&timeSyncInterval=60&runAsServer=off

Response data:
returnCode=0

4. System Management

4.1. Factory Default

Reload factory defaults.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	systemMgmt	
mode	Mode Code	702	
withoutItems	Exception item	network ex> withoutItems=network	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=systemMgmt&mode=702&withoutItems=network

Response data:
returnCode=0

4.2. Restart

Restart system.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	systemMgmt	
mode	Mode Code	9000	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=systemMgmt&mode=9000

Response data:

returnCode=0

4.3. Firmware Upgrade (for model group 1 & model group 2 only)

Upgrade system. About 70-100 seconds will be taken for response.
Make sure that preparing for File Upload is preceded. Please refer Preparing File Upload.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?action=firmwareUpgrade

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	firmwareUpgrade	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	0: Upgrade success 101: Upgrade fail Refer Return Code	

Example

Send data:
action=firmwareUpgrade

Response data:
returnCode=0

4.4. Setup Import

Imports selected setup file to the camera.
Make sure that preparing for File Upload is preceded. Please refer Preparing File Upload.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	importSetup	
mode	Mode Code	9001	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=importSetup&mode=9001

Response data:
returnCode=0

4.5. **Setup Export**

Exports camera setup file from the camera.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	exportSetup	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	Group1,2
dataFileURL			Group1,2
	Xml File		Group3,4,5

Example for Group1,2

Send data:
action= exportSetup&mode=1

Response data:
returnCode=0&dataFileURL=http://10.0.126.70/registry.dat

Example for Group3,4,5

Send data:
action= exportSetup&mode=1

Response data:

```
<?xml version="1.0" encoding="utf-8" ?>
<ConfigurationSet version="1" type="object" key="1">
        <ConfigurationCategory type="object" key="255">
                <SubCategoryArray type="array" key="3">
                        <SubCategory type="object" key="2">
…
…
        </OnvifSetup>
<Hash type="string" method="md5">68A4A6B5D4DB6E64E9D84C69F205AE0B</Hash>
</ConfigurationSet>
```

4.6. **Read storage information**

Get storage information

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	storage	
mode	Mode Code	1	
type	storage command	information ex>type=information	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
capacity	storage capacity (Bytes)	<integer> ex>capacity=1000000000	

Example

Send data:

action=storage&mode=1&type=information

Response data:

returnCode=0&capacity=1000000000

4.7. Format storage

Format storage

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	storage	
mode	Mode Code	0	
type	storage command	format ex>type=format	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=storage&mode=0&type=format

Response data:

returnCode=0

4.8. Camera Model Information (for Setup Style)

Get camera model information

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	modelInformation	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
model	Device model name	<string> ex> model=NC-B320IP	
modelGroup	Setup method group	<integer> ex>modelGroup=1	
mac	MAC address	<string> ex>mac=00:03:22:12:34:56	
softwareVersion	SW Version	<string> ex> softwareVersion=1.0	
hardwareVersion	HW Version	<string> ex> hardwareVersion=1.0	
resolutionList	supported resolution list	<string> ex> resolutionList=1920x1080|1280x960|704x480	
webApiVersion	Web API Version	<string> ex> webApiVersion=1.8	

Example

Send data:

action=modelInformation&mode=1

Response data:

returnCode=0&model=NC-B320IP&modelGroup=1&mac=00:03:22:12:34:56&softwareVersion=1.0&hardwareVersion=1.0&webApiVersion=1.7

4.9. **Set Camera Encoding Mode**

Set camera encoding mode

Method: POST

Syntax
http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	cameraMode	
mode	Mode Code	0	
type	Camera mode type	30ips | 60ips ex>type=30ips	legacy
ipsMode	Camera ips mode	30ips | 60ips ex>ipsMode=30ips	
resolutionMode	Camera resolution mode	2M | 3M ex>resolutionMode=2M	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=cameraMode&mode=0&ipsMode=30ips&resolutionMode=2M

Response data:
returnCode=0

4.10. Get Camera Encoding mode

Get camera encoding mode

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	cameraMode	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
type	Camera mode type	30ips | 60ips ex>type=30ips	legacy
ipsMode	Camera ips mode	30ips | 60ips ex>ipsMode=30ips	
resolutionMode	Camera resolution mode	2M | 3M ex>resolutionMode=2M	

Example

Send data:
action=cameraMode&mode=1

Response data:
returnCode=0&type=30ips&ipsMode=30ips&resolutionMode=2M

4.11. Set Alarm out

Set camera alarm out manually

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	manualAlarmOut	
mode	Mode Code	0	
alarmOut	manual alarm out trigger	on | off ex> output=on	Group 1,2
active	manual alarm out trigger	on | off ex> active=on	Group 3~6
index	Set alarm out index on cameras that support multiple alarm out.	ex> index=1	Group 3~6 & Supports multiple alarm outs

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=manualAlarmOut &mode=0&output=on

Response data:

returnCode=0

4.12. Get Public Key (except model group 1 & model group 2)

Get RSA public key.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	getPublicKey	
Mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
n	RSA N	<hexadecimal string> ex> n=A5BBF31B17254A9F37F91FDEC5326DB5C0F843B549A 6FB38B4754D8F2325D05D9E57D9AEB2D666DA030A05C 844F4289C1DD2D9134B98A171E782EDBD833E8A1A250D B1C7BB52DDDD9DB90F02A8823D444E443BC47F5623A0 44F8E6DDA0851D9A90D0DCFC304C47462E9F65DF3DAB EB8B8F504911271A07AC9CE294FD04B	
e	RSA E	<integer> ex> e=3	

Example

Send data:

action=initialize&userName=admin&userPassword=passwd&userSms=1234567890&userEmail=jone@jone.com

Response data:

returnCode=0&n=A5BBF31B17254A9F37F91FDEC5326DB5C0F843B549A36FB38B4754D8F2325D05D9E57D9AEB2D666DA030A05C7844F4289C1DD2D9134B98A171E782EDBD833E8A1A250D7B1C7BB52DDDD9DB90F02A8823D444E443BC47F5623A0F44F8E6DDA0851D9A90D0DCFC304C47462E9F65DF3DAB3EB8B8F504911271A07AC9CE294FD04B&e=3

4.13. Initialize user (except model group 1 & model group 2)

Operates only within KOR OEM and factory default state.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	initialize	
Mode	Mode Code	0	
userName	User name	<string> ex> userName=admin	
userPassword	User password (for password reset)	< hexadecimal string encypted By RSA public Key> ex> userPassword=30a8dcc18f9f7dad53381ff7bfe561b0f0fc00f9d2d5b5dbb4066ae448369321ad1d7d3b2ea157bca8e8787b9ec79c81d07fa8230ff992668a6e262fc1e4261d86a15d3b449760371ff0376a42b351d8203dae4c5f204ec8a48b06e34ff9ed98ecf2c7c192d380277214be4e9f3df8293cf29d6ab0938a96248f907c965fb25e	Strong Password
userSms	User SMS (for password reset)	< hexadecimal string encypted By RSA public Key > ex> userSms=670fe4899fdb5e24af1d4438a71d27dbc106970df4ba802a7504ca05a42581eb0102d329c90634d609c3e5dd65326e5bb944bd9cd4d2513cbec6373f11cae313d157cfeb3ae19fdccb04dc653d5aec433f4a255cfd1879bfc06b3d9ef1bc214976732e418e749c11cdbed9a1522abd26219ec7e3c4c8a7354e2b9aa646672cca	
userEmail	User e-mail (for password reset)	< hexadecimal string encypted By RSA public Key > ex> userEmail=8fbdc81476b38074d4deb4abb68604a139492cbf16a112686d763f625873a161ab69a231a19ff745e8ee5a04bc67e12a822b12161f1a1e9185483e46b97cb0157a3cdab9f93a7045e9c00e8e5adf9b0dd1cc8bc35bf5221fe8a0aa4133aedaea25cd25d191c9e722da06965a45ead631f6d7d08c8f7613d973b9bcf7e3281815	
userCountry	User Country Code (for password reset) ex> 82 is for korea	< hexadecimal string encypted By RSA public Key > ex> userCountry=670fe4899fdb5e24af1d4438a71d27dbc106970df4ba802a7504ca05a42581eb0102d329c90634d609c3e5dd65326e5bb944bd9cd4d2513cbec6373f11cae313d157cfeb3ae19fdccb04dc653d5aec433f4a255cfd1879bfc06b3d9ef1bc214976732e418e749c11cdbed9a1522abd26219ec7e3c4c8a7354e2b9aa646672cca	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=initialize&userName=admin&userPassword=30a8dcc18f9f7dad53381ff7bfe561b0f0fc00f9d2d5b5dbb4066ae448369321ad1d

7d3b2ea157bca8e8787b9ec79c81d07fa8230ff992668a6e262fc1e4261d86a15d3b449760371ff0376a42b351d8203dae4c5f204ec8a
48b06e34ff9ed98ecf2c7c192d380277214be4e9f3df8293cf29d6ab0938a96248f907c965fb25e&userEmail=670fe4899fdb5e24af1d44
38a71d27dbc106970df4ba802a7504ca05a42581eb0102d329c90634d609c3e5dd65326e5bb944bd9cd4d2513cbec6373f11cae313d
157cfeb3ae19fdccb04dc653d5aec433f4a255cfd1879bfc06b3d9ef1bc214976732e418e749c11cdbed9a1522abd26219ec7e3c4c8a73
54e2b9aa646672cca&userSms=8fbdc81476b38074d4deb4abb68604a139492cbf16a112686d763f625873a161ab69a231a19ff745e8
ee5a04bc67e12a822b12161f1a1e9185483e46b97cb0157a3cdab9f93a7045e9c00e8e5adf9b0dd1cc8bc35bf5221fe8a0aa4133aeda
ea25cd25d191c9e722da06965a45ead631f6d7d08c8f7613d973b9bcf7e3281815&userCountry=670fe4899fdb5e24af1d4438a71d27db
c106970df4ba802a7504ca05a42581eb0102d329c90634d609c3e5dd65326e5bb944bd9cd4d2513cbec6373f11cae313d157cfeb3ae1
9fdccb04dc653d5aec433f4a255cfd1879bfc06b3d9ef1bc214976732e418e749c11cdbed9a1522abd26219ec7e3c4c8a7354e2b9aa64
6672cca&mode=0

Response data:
returnCode=0

5. User Management

5.1. Read

Get user accounts.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	userSetup	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
userCount	Number of registered users	<integer> ex> userCount=2	
user{N}Name	User ID	<string> ex> user1Name=admin	
user{N}Group	User group	Administrator | Operator | User | <user defined group> ex> user1Group=Administrator	
{N} : 1 ~ userCount			

Example

Send data:

action=userSetup&mode=1

Response data:

returnCode=0&userCount=2&user1Name=admin&user1Group=Administrator&user2Name=usr1&user2Group=User

5.2. **Write (Secured)**

Set user accounts. This command can be only used with https or RSA encryption.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	userSetup	
mode	Mode Code	0 | 5 ex> mode=0	
userWriteMode	Set mode	edit | add | remove ex> userWriteMode=add	
userName	User ID	<string> ex> userName=jone	
userPassword	User password	If the mode is 0, then this command can be only used with https. <string> If remove mode, it does not need. ex> userPassword =passwd	Strong Password for KOR OEM
		If mode is 5 < hexadecimal string encypted By RSA public Key> If remove mode, it does not need. ex> userPassword=30a8dcc18f9f7dad53381ff7bfe561b0f0fc00f9d2d5b5dbb4066ae448369321ad1d7d3b2ea157bca8e8787b9ec79c81d07fa8230ff992668a6e262fc1e4261d86a15d3b449760371ff0376a42b351d8203dae4c5f204ec8a48b06e34ff9ed98ecf2c7c192d380277214be4e9f3df8293cf29d6ab0938a96248f907c965fb25e	
userEmail	User Email (for password reset)	If the mode is 0, then this command can be only used with https. <string> If remove mode, it does not need. ex> userEmail =jone@jone.com	KOR OEM only
		If mode is 5 < hexadecimal string encypted By RSA public Key> If remove mode, it does not need. ex> userEmail=8fbdc81476b38074d4deb4abb68604a139492cbf16a112686d763f625873a161ab69a231a19ff745e8ee5a04bc67e12a822b12161f1a1e9185483e46b97cb0157a3cdab9f93a7045e9c00e8e5adf9b0dd1cc8bc35bf5221fe8a0aa4133aedaea25cd25d191c9e722da06965a45ead631f6d7d08c8f7613d973b9bcf7e3281815	
userSms	User SMS number (for password reset)	If the mode is 0, then this command can be only used with https. <string> If remove mode, it does not need. ex> userSms =1234567890	KOR OEM only
		If mode is 5 < hexadecimal string encypted By RSA public Key> If remove mode, it does not need. ex> userSms=670fe4899fdb5e24af1d4438a71d27dbc106970df4ba	

		802a7504ca05a42581eb0102d329c90634d609c3e5dd65326e5 bb944bd9cd4d2513cbec6373f11cae313d157cfeb3ae19fdccb04 dc653d5aec433f4a255cfd1879bfc06b3d9ef1bc214976732e418 e749c11cdbed9a1522abd26219ec7e3c4c8a7354e2b9aa64667 2cca	
userCountry	User Country Code (for password reset) ex> 82 is for korea	If the mode is 0, then this command can be only used with https. <string> If remove mode, it does not need. ex> userCountry =1234567890	
		If mode is 5 < hexadecimal string encypted By [RSA public Key](#)> If remove mode, it does not need. ex> userCountry=670fe4899fdb5e24af1d4438a71d27dbc106970df4 ba802a7504ca05a42581eb0102d329c90634d609c3e5dd65326 e5bb944bd9cd4d2513cbec6373f11cae313d157cfeb3ae19fdccb 04dc653d5aec433f4a255cfd1879bfc06b3d9ef1bc214976732e4 18e749c11cdbed9a1522abd26219ec7e3c4c8a7354e2b9aa646 672cca	
userGroup	User group	Administrator | Operator | User | <user defined group> If remove mode, it does not need. ex> userGroup=User	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer [Return Code](#)	

Example

Send data:

action=userSetup&mode=0&userWriteMode=add&userName=jone&userPassword=passwd&userEmail=jone@jone.com&userSms=1234567890&userGroup=User&userCountry=82

Response data:

returnCode=0

6. Group Management

6.1. Read

Get defined user groups..

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	groupSetup	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
allowAnonymousLogin	allow anonymous login	on | off	
allowAnoniymousPTZ	allow anonymous PTZ control	on | off	
groupCount	Number of registered group	<integer> ex> groupCount=2	
groupName{N}	Group Name	<string> ex> groupName1=testGroup	
authorities{N}	authorities for registerd group	<string> upgrade | setup | color | ptz | alarmOut | search | clipCopy | systemCheck ex> authorities1=upgrade|setup	
{N} : 1 ~ groupCount			

Example

Send data:

action=groupSetup&mode=1

Response data:

returnCode=0&groupCount=3&groupName1=Administrator&authorities1=upgrade|setup|color|ptz|alarmOut

6.2. **Write**

Edit/Add/Remove of user group.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	**Description**	**Values**	**ETC**
action	Action ID	groupSetup	
mode	Mode Code	0	
allowAnonymousLogin	allow anonymous login	on | off	
allowAnoniymousPTZ	allow anonymous PTZ control	on | off	
groupWriteMode	Set mode	edit | add | remove ex> groupWriteMode =add	
groupName	Group name	<string> ex> groupName = tester	
authorities	authorities for registerd group	upgrade | setup | color | ptz | alarmOut | search | clipCopy | systemCheck If remove mode, it does not need. ex> authorities1=upgrade|setup	

Receive Parameters

Parameters	**Description**	**Values**	**ETC**
returnCode	Return code	Refer Return Code	

Example

Send data:
action=groupSetup&mode=0&userWriteMode=add&groupName=tester&authorities=upgrade|setup|color|ptz|alarmOut

Response data:
returnCode=0

7. Network – IP Address

7.1. Read

Get network address information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkIp	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
type	Boot protocol type	manual | dhcp | adsl ex>type=manual	
ipAddress	IP address	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> ipAddress=192.168.1.110	
gateway	Gateway address	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> gateway=192.168.1.1	
subnetMask	Subnet mask	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> subnetMask=225.255.255.0	
dnsServer	DNS server address	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> dnsServer=192.168.0.254	
dnsServerFromDHCP	Get DNS server from DHCP	on | off ex> dnsServerFromDHCP=on	
adslID	user ID when use ADSL	<string> ex>adslID=usersapmple	Group1,2
adslPassword	uuse password when use ADSL	<string> ex>adslPassword=password	Group1,2
linkLocalOnly	Link-local Only	on | off ex> linkLocalOnly=off	
videoMulticastAddress	Video multicast address	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> videoMulticastAddress=235.235.235.235	
videoMulticastPort	Video multicast port	<port> ex> videoMulticastPort=38016	
audioMulticastAddress	Audio multicast address	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> audioMulticastAddress=235.235.235.235	Only when audio-supported.
audioMulticastPort	Audio multicast port	<port> ex> audioMulticastPort=38016	Only when audio-supported.

Example

Send data:
action=networkIp&mode=1

Response data:
returnCode=0&type=manual&ipAddress=192.168.0.250&gateway=192.168.0.254&subnetMask=225.255.255.0&dnsServer=192.168.0.254&dnsServerFromDHCP=off&adslID=usersapmple&adslPassword=password&linkLocalOnly=off&videoMulticastAddress=235.235.235.235&videoMulticastPort=38016&audioMulticastAddress=235.235.235.235&audioMulticastPort=38016

7.2. Write (Secured)

Set network address information. This command can be only used with https or RSA encryption.

Method: POST

Syntax

https://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkIp	
mode	Mode Code	0 | 5 ex> mode=0	
type	Boot protocol type	manual | dhcp | adsl ex>type=manual	
ipAddress	IP address	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> ipAddress=192.168.1.110	
gateway	Gateway address	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> gateway=192.168.1.1	
subnetMask	Subnet mask	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> subnetMask=225.255.255.0	
dnsServer	DNS server address	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> dnsServer=192.168.0.254	
dnsServerFromDHCP	Get DNS server from DHCP	on | off ex> dnsServerFromDHCP=on	
adslID	user ID when use ADSL	<string> ex>adslID=usersapmple	Group1,2
adslPassword	uuse password when use ADSL	If the mode is 0, then this command can be only used with https. <string> ex>adslPassword=password	Group1,2
		If mode is 5 < hexadecimal string encypted By RSA public Key> ex> adslPassword=30a8dcc18f9f7dad53381ff7bfe561b0f0fc00f9d2d5b5dbb4066ae448369321ad1d7d3b2ea157bca8e8787b9ec79c81d07fa8230ff992668a6e262fc1e4261d86a15d3b449760371ff0376a42b351d8203dae4c5f204ec8a48b06e34ff9ed98ecf2c7c192d380277214be4e9f3df8293cf29d6ab0938a96248f907c965fb25e	
linkLocalOnly	Link-local Only	on | off ex> linkLocalOnly=off	
videoMulticastAddress	Video multicast address	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> videoMulticastAddress=235.235.235.235	
videoMulticastPort	Video multicast port	<port>	

		ex> videoMulticastPort=38016	
audioMulticastAddress	Audio multicast address	<IPv4 address> 0.0.0.0 ~ 255.255.255.255 ex> audioMulticastAddress=235.235.235.235	Only when audio-supported.
audioMulticastPort	Audio multicast port	<port> ex> audioMulticastPort=38016	Only when audio-supported.

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=networkIp&mode=0&type=manual&ipAddress=192.168.1.110&gateway=192.168.1.1&subnetMask=225.255.255.0&dnsServer=192.168.0.254&dnsServerFromDHCP=off&adslID=usersapmple&adslPassword=password&linkLocalOnly=off&videoMulticastAddress=235.235.235.235&videoMulticastPort=38016&audioMulticastAddress=235.235.235.235&audioMulticastPort=38016

Response data:

returnCode=0

8. Network – IPv6 Address

8.1. Read

Get network IPv6 address information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkIPv6	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
ipv6Address	IP address list	<IPv6 address> <IPv6 address>|<IPv6 address> ex> 2001:0db8::121|fe80::	Read only
useIpv6	Use ipv6 address	on | off ex> useIpv6=on	
type	Boot protocol type	manual | dhcp ex>type=manual	
ipAddress	IP address	<IPv6 address> 0:0:0:0:0:0:0:0 ~ ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff ex> ipAddress= 2001:0db8::121	
gateway	Gateway address	<IPv6 address> 0:0:0:0:0:0:0:0 ~ ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff ex> gateway=2001:0db8::1	
prefix	Prefix	0~64 ex> 64	
dnsServer	DNS server address	<IPv6 address> 0:0:0:0:0:0:0:0 ~ ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff ex> dnsServer=2001:0db8::8888	
dnsServerFromDHCP	Get DNS server from DHCP	on | off ex> dnsServerFromDHCP=off	

Example

Send data:

action=networkIpv6&mode=1

Response data:

returnCode=0&ipv6Address=2001:db8::7334|fe80::&useIpv6=on&ipAddress=2001:0db8::7334&type=manual&
gateway=2001:0db8::1&prefix=64&dnsServerFromDHCP=off&dnsServer=2001:0db8::8888

8.2. Write

Set network IPv6 address information.

Method: POST

Syntax

https://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkIPv6	
useIpv6	Use ipv6 address	on | off ex> useIpv6=on	
type	Boot protocol type	manual | dhcp ex>type=manual	
ipAddress	IP address	<IPv6 address> 0:0:0:0:0:0:0:0 ~ ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff ex> ipAddress= 2001:0db8::121	
gateway	Gateway address	<IPv6 address> 0:0:0:0:0:0:0:0 ~ ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff ex> gateway=2001:0db8::1	
prefix	Prefix	0~64 ex> 64	
dnsServer	DNS server address	<IPv6 address> 0:0:0:0:0:0:0:0 ~ ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff ex> dnsServer=2001:0db8::8888	
dnsServerFromDHCP	Get DNS server from DHCP	on | off ex> dnsServerFromDHCP=off	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=networkIPv6&mode=0&useIpv6=on&type=manual&ipAddress=2001:0db8::7334&gateway=2001:0db8::1&prefix=64&dnsServer=2001:0db8::8888&dnsServerFromDHCP=off

Response data:

returnCode=0

9. Network – DDNS

9.1. Read

Get DDNS information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkDDNS	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useDDNS	Use DDNS on/off	on | off ex> useDDNS=on	
serverAddress	DDNS server address	<string> ex> ipAddress=dvrnames.net	
port	DDNS server port	<integer> 10000 ~ 12000 ex> port=10088	
useNAT	Use NAT on/off	on | off ex> useNAT=off	
cameraName	System name to register DDNS server	<string> ex> cameraName=dvrnssample	

Example

Send data:

action=networkDDNS&mode=1

Response data:

returnCode=0&use DDNS =on&serverAddress=dvrnames.net&port=10088&useNAT=off&cameraName=dvrnssample

9.2. Write

Set DDNS information.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	network DDNS	
mode	Mode Code	0	
useDDNS	Use DDNS on/off	on | off ex> useDDNS=on	
serverAddress	DDNS server address	<string> ex> ipAddress=dvrnames.net	
port	DDNS server port	<integer> ex> port=10088	
useNAT	Use NAT on/off	on | off ex> useNAT=off	
cameraName	System name to register DDNS server	<string> ex> cameraName=dvrnssample	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=networkDDNS&mode=0&useDDNS=on&serverAddress=dvrnames.net&port=10088&useNAT=off&cameraName=dvrnssample

Response data:

returnCode=0

9.3. Check DDNS

test DDNS service

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	network DDNS	
mode	Mode Code	2	
serverAddress	DDNS server address	<string> ex> ipAddress=dvrnames.net	
port	DDNS server port	<integer> ex> port=10088	
useNAT	Use NAT on/off	on | off ex> useNAT=off	
cameraName	System name to register DDNS server	<string> ex> cameraName=dvrnssample	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code return 0 when success, 110 means DDNS test failed	

Example

Send data:

action=networkDDNS&mode=2&serverAddress=dvrnames.net&port=10088&useNAT=off&cameraName=dvrnssample

Response data:

returnCode=0

10. Network – Port

10.1. Read

Get network service ports.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkPort	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
adminPort	Remote control port	<integer> ex>adminPort=8200	Group 1,2 only
adminDSCP	Remote control QoS	<integer> ex> adminDSCP=0	Group 1,2 only
watchPort	Live video port	<integer> ex> watchPort=8016	Group 1,2 only
watchDSCP	Live video QoS	<integer> ex> watchDSCP=0	Group 1,2 only
recordPort	Recording video port	<integer> ex> recordPort=8017	Group 1,2 only
recordDSCP	Recording video QoS	<integer> ex> recordDSCP=0	Group 1,2 only
searchPort	Playback port	<integer> ex> searchPort=10019	Group 1,2 only
searchDSCP	Playback QoS	<integer> ex> searchDSCP=0	Group 1,2 only
remotePort	Playback port	<integer> ex> remotePort=10019	
remoteDSCP	Playback QoS	<integer> ex> remoteDSCP=0	
useWeb	Use web service on/off	on | off ex> useWeb=on	
webPort	Web service port	<integer> ex> webPort=80	
webDSCP	Web service QoS	<integer> ex> webDSCP=0	
useRtsp	Use RTSP on/off	on | off ex> useRtsp=on	
rtspPort	RTSP port	<integer> ex> rtspPort=554	
rtspDSCP	RTSP QoS	<integer> ex> rtspDSCP=0	
useHTTPS	Use HTTPS on/off	on | off ex> useHTTPS=off	

useUPNP	Use UPNP on/off	on | off ex> useUPNP=off	

Example

Send data:

action=networkPort&mode=1

Response data:

returnCode=0&adminPort=8200&adminDSCP=0&watchPort=8016&watchDSCP=0&recordPort=8017&recordDSCP=0&searchPort=10019&searchDSCP=0&useWeb=on&webPort=80&webDSCP=0&useRTSP=on&rtspPort=554&rtspDSCP=0&useHTTPS=off&useUPNP=on

10.2. **Write**

Set network service ports.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkPort	
mode	[Mode Code]()	0	
adminPort	Remote control port	<integer> ex>adminPort=8200	Group 1,2 only
adminDSCP	Remote control QoS	<integer> ex> adminDSCP=0	Group 1,2 only
watchPort	Live video port	<integer> ex> watchPort=8016	Group 1,2 only
watchDSCP	Live video QoS	<integer> ex> watchDSCP=0	Group 1,2 only
recordPort	Recording video port	<integer> ex> recordPort=8017	Group 1,2 only
recordDSCP	Recording video QoS	<integer> ex> recordDSCP=0	Group 1,2 only
searchPort	Playback port	<integer> ex> searchPort=10019	Group 1,2 only
serachDSCP	Playback QoS	<integer> ex> searchDSCP=0	Group 1,2 only
remotePort	remote port	<integer> ex> remotePort=10019	
remoteDSCP	remote QoS	<integer> ex> remoteDSCP=0	
useWeb	Use web service on/off	on | off ex> useWeb=on	
webPort	Web service port	<integer> ex> webPort=80	
webDSCP	Web service QoS	<integer> ex> webDSCP=0	
useRtsp	Use RTSP on/off	on | off ex> useRtsp=on	
rtspPort	RTSP port	<integer> ex> rtspPort=554	
rtspDSCP	RTSP QoS	<integer> ex> rtspDSCP=0	
useHTTPS	Use HTTPS on/off	on | off ex> useHTTPS=off	
useUPNP	Use UPNP on/off	on | off ex> useUPNP=off	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=networkPort&mode=0&adminPort=8200&adminDSCP=0&watchPort=8016&watchDSCP=0&recordPort=8017&recordDSCP=0&searchPort=10019&searchDSCP=0&useWeb=on&webPort=80&webDSCP=0&useRTSP=on&rtspPort=554&rtspDSCP=0&useHTTPS=off&useUPNP=on

Response data:

returnCode=0

10.3. check UPNP

check UPNP for network service ports.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkPort	
mode	Mode Code	2	
adminPort	Remote control port	<integer> ex>adminPort=8200	
watchPort	Live video port	<integer> ex> watchPort=8016	
recordPort	Recording video port	<integer> ex> recordPort=8017	
searchPort	Playback port	<integer> ex> searchPort=10019	
remotePort	remote port	<integer> ex> remotePort=10019	Group3
webPort	Web service port	<integer> ex> webPort=80	
rtspPort	RTSP port	<integer> ex> rtspPort=554	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
conflicPort	changed Port List	admin | watch | record | search | web | rtsp ex>conflicPort=admin | web	
adminPort	Remote control port	<integer> ex>adminPort=8200	
watchPort	Live video port	<integer> ex> watchPort=8016	
recordPort	Recording video port	<integer> ex> recordPort=8017	
searchPort	Playback port	<integer> ex> searchPort=10019	
remotePort	remote port	<integer> ex> remotePort=10019	Group3
webPort	Web service port	<integer> ex> webPort=80	
rtspPort	RTSP port	<integer> ex> rtspPort=554	

Example

Send data:

action=networkPort&mode=2&adminPort=8200&watchPort=8016&recordPort=8017&searchPort=10019&webPort=80&rtspPort=554

Response data:
returnCode=0&conflicPort=admin|web&adminPort=8201&webPort=81

11. Network – Bandwidth

11.1. Read

Get network bandwidth control.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkBandwidth	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useNetworkBandwidth	Use bandwidth limit on/off	on | off ex> useNetworkBandwidth=off	
networkBandwidth	Bandwidth (Kbps)	1~102400 ex> networkBandwidth=102400	

Example

Send data:
action=networkBandwidth&mode=1

Response data:
returnCode=0&useNetworkBandwidth=on&networkBandwidth=102400

11.2. Write

Get network bandwidth control.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkBandwidth	
mode	Mode Code	0	
useNetworkBandwidth	Use bandwidth limit on/off	on | off ex> useNetworkBandwidth=off	
networkBandwidth	Bandwidth (Kbps)	1~102400 ex> networkBandwidth=102400	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=networkBandwidth&mode=0&useNetworkBandwidth=on&networkBandwidth=102400

Response data:
returnCode=0

12. Network – Security

12.1. Read

Get network security information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkSecurity	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
filterType	IP filtering strategy	off | allow | deny ex> filterType=off	
allowList	IP allow list	<IPv4 address> | <IPv4 address>_<IPv4 address> max 32 items ex> 192.168.10.200|192.168.20.1_192.168.20.255	
denyList	IP deny list	<IPv4 address> | <IPv4 address>_<IPv4 address> max 32 items ex> 192.168.10.200|192.168.20.1_192.168.20.255	
useSSL	Use SSL on/off	on | off ex> useSSL=off	
sslType	SSL strength type	standard | high ex> sslType=high	

Example

Send data:
action=networkSecurity&mode=1

Response data:
returnCode=0&filterType=deny&allowList=&denyList=192.168.10.255|192.168.11.1_192.168.11.255&useSSL=off&sslType=high

12.2. **Write**

Set network security information.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	**Description**	**Values**	**ETC**
action	Action ID	networkSecurity	
mode	Mode Code	0	
filterType	IP filtering strategy	off | allow | deny ex> filterType=off	
allowList	IP allow list	<IPv4 address> | <IPv4 address>_<IPv4 address> max 32 items ex> 192.168.10.200|192.168.20.1_192.168.20.255	
denyList	IP deny list	<IPv4 address> | <IPv4 address>_<IPv4 address> max 32 items ex> 192.168.10.200|192.168.20.1_192.168.20.255	
useSSL	Use SSL on/off	on | off ex> useSSL=off	
sslType	SSL strength type	standard | high ex> sslType=high	

Receive Parameters

Parameters	**Description**	**Values**	**ETC**
returnCode	Return code	Refer Return Code	

Example

Send data:

action=networkSecurity&mode=0&filterType=deny&allowList=&denyList=192.168.10.255|192.168.11.1_192.168.11.255&useSSL=off
&sslType=high

Response data:

returnCode=0

13. Network – IEEE 802.1X

13.1. Read

Get network IEEE 802.1X information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkIEEE802_1X	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useIEEE802_1X	use IEEE802.1X	on | off ex> filterType=off	
EAPType	type of EAP	MD5 | TLS | PEAP ex> EAPType=MD5	
EAPOLVersion	version of EAP	1 | 2 ex> EAPOLVersion=1	
EAPIdentify	ID for EAP	<string>	
caCertificate	CA Certificate file name (need file when EAP type TLS or PEAL)	<string> <filename> | fileNotFound ex> caCertificate=fileNotFound	
clientCertificate	Client Certificate file name (need file when EAP type TLS)	<string> <filename> | fileNotFound ex> clientCertificate=fileNotFound	
clientPrivateKey	Client Private Key file name (need file when EAP type TLS)	<string> <filename> | fileNotFound ex> clientPrivateKey=fileNotFound	

Example

Send data:
action=networkIEEE802_1X&mode=1

Response data:
returnCode=0&useIEEE802_1X=on&EAPType=MD5&EAPOLVersion=1&EAPIdentify=&caCertificate=fileNotFound&clientCertificate=fileNotFound&clientPrivateKey=fileNotFound

13.2. **Write (Secured)**

Get network security information. This command can be only used with https or RSA encryption.

Method: POST

Syntax

https://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkIEEE802_1X	
mode	Mode Code	0 | 5 ex> mode=0	
useIEEE802_1X	use IEEE802.1X	on | off ex> useIEEE802_1X=off	
EAPType	type of EAP	MD5 | TLS | PEAP ex> EAPType=MD5	
EAPOLVersion	version of EAP	1 | 2 ex>EAPOLVersion=1	
EAPIdentify	ID for EAP	<string> ex>EAPIdentify=tester	
EAPPassword	password for EAP (need when EAP type is MD5 or PEAP)	If the mode is 0, then this command can be only used with https. <string> ex>EAPPassword=1234	
		If mode is 5 < hexadecimal string encypted By RSA public Key> ex> EAPPassword=30a8dcc18f9f7dad53381ff7bfe561b0f0fc00f9d2d5b5dbb4066ae448369321ad1d7d3b2ea157bca8e8787b9ec79c81d07fa8230ff992668a6e262fc1e4261d86a15d3b449760371ff0376a42b351d8203dae4c5f204ec8a48b06e34ff9ed98ecf2c7c192d380277214be4e9f3df8293cf29d6ab0938a96248f907c965fb25e	
privateKeyPassword	password for client private key (need when EAP type is TLS)	If the mode is 0, then this command can be only used with https. <string> ex>privateKeyPassword=password	
		If mode is 5 < hexadecimal string encypted By RSA public Key> ex> privateKeyPassword=30a8dcc18f9f7dad53381ff7bfe561b0f0fc00f9d2d5b5dbb4066ae448369321ad1d7d3b2ea157bca8e8787b9ec79c81d07fa8230ff992668a6e262fc1e4261d86a15d3b449760371ff0376a42b351d8203dae4c5f204ec8a48b06e34ff9ed98ecf2c7c192d380277214be4e9f3df8293cf29d6ab0938a96248f907c965fb25e	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=networkIEEE802_1X&mode=0&useIEEE802_1X=on&EAPType=MD5&EAPOLVersion=1&EAPIdentify=tester&EAPPassword=1234

Response data:
returnCode=0

13.3. Upload file

Upload certification files for IEEE 802.1X. This command can be only used with https.
Make sure that preparing for File Upload is preceded. Please refer Preparing File Upload.

Method: POST

Syntax

https://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkIEEE802_1X	
mode	Mode Code	9001	
type	file type for current upload	caCertificate | clientCertificate | clientPirvateKey	
fileName	audio file name for upload	<string> ex> filename=privateKey	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action= networkIEEE802_1X&mode=9001&type=clientPirvateKey&filename=privateKey

Response data:
returnCode=0

14. Network– SIP

14.1. Read

Get SIP information

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkSIP	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useSip	Use SIP on/off	on | off ex> useSip=on	
sipServer	SIP server address	<IPv4 address> ex> sipServer=10.0.126.70	
port	SIP server port	<integer> ex>port=5060	
userID	SIP user ID	<string> ex>userID=user	
password	SIP user password	<string> ex>password=passwd	
useSipVideo	Use SIP Video Streaming	on | off ex> useSipVideo=on	
sipVideoStream	SIP Video Stream channel	primary | secondary | tertiary ex> sipVideoStream=primary	

Example

Send data:
action=actionFtp&mode=1

Response data:
returnCode=0&useSip=on&sipServer=10.0.126.70&port=5060&userID=user&password=pass&useSipVideo=on&sipVideoStream=primary

14.2. **Write**

Set SIP information for event action.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	networkSIP	
mode	[Mode Code](#)	0	
useSip	Use SIP on/off	on | off ex> useSip=on	
sipServer	SPI server address	<IPv4 address> ex> sipServer=10.0.126.70	
port	SIP server port	<integer> ex>port=5060	
userID	SIP user ID	<string> ex>userID=user	
password	SIP user password	<string> ex>passwrd=passwd	
useSipVideo	Use SIP Video Streaming	on | off ex> useSipVideo=on	
sipVideoStream	SIP Video Stream channel	primary | secondary | tertiary ex> sipVideoStream=primary	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer [Return Code](#)	

Example

Send data:

action=actionFtp&mode=0&useFtp=on&ftpServer1=10.0.126.70&uploadPath1=event&port1=21&userID1=user&password1=passwd&uploadType=event&uploadFrequency=1m&resolution=352x240&quality=basic&prefix=image&namingType=datetime

Response data:

returnCode=0

15. Video – Easy Video Setting(Self adjust video mode)

15.1. Read

Get easy video setting(Self adjust video mode) information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoEasySetting	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
easyDayType	Easy setting Daytime mode	0 ~ 3 ex> easyDayType=0 0 : Custom 1 : Natural 2 : Vivid 3 : Denoise	Easy video setting support model
easyDaySharpness	Easy setting Daytime sharpness	0 ~ 3 (Can be set only in easy day type custom mode.) ex> easyDaySharpness=1	
easyDayContrast	Easy setting Daytime contrast	0 ~ 2 (Can be set only in easy day type custom mode.) ex> easyDayContrast=2	
easyDayColors	Easy setting Daytime Colors	0 ~ 2 (Can be set only in easy day type custom mode.) ex> easyDayColors=1	
easyDayBrightness	Easy setting Daytime Brightness	0 ~ 2 (Can be set only in easy day type custom mode.) ex> easyDayBrightness=2	
easyNightType	Easy setting Nighttime mode	0 ~ 3 ex> easyNightType=0 0 : Custom 1 : Natural 2 : Vivid 3 : Denoise	
easyNightSharpness	Easy setting Nighttime sharpness	0 ~ 3 (Can be set only in easy night type custom mode.) ex> easyNightSharpness=3	
easyNightGamma	Easy setting Nighttime contrast	0 ~ 2 (Can be set only in easy night type custom mode.) ex> easyNightGamma=2	
easyNightBrightness	Easy setting Nighttime brightness	0~2 (Can be set only in easy night type custom mode.) ex> easyNightBrightness=1	

Example

Send data:
action=videoEasySetting&mode=1

Response data:
returnCode=0&easyDayType=0&easyDaySharpness=0&easyDayContrast=0&easyDayColors=0&easyDayBrightness=0&easyNightType=0&easyNightSharpness=0&easyNightGamma=0&easyNightBrightness=0

15.2. **Write**

Set video-image information.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	**Description**	**Values**	**ETC**
action	Action ID	videoEasySetting	
easyDayType	Easy setting Daytime mode	0 ~ 3 ex> easyDayType=0 0 : Custom 1 : Natural 2 : Vivid 3 : Denoise	Easy video setting support model
easyDaySharpness	Easy setting Daytime sharpnes	0 ~ 3 (Can be set only in easy day type custom mode.) ex> easyDaySharpness=1	
easyDayContrast	Easy setting Daytime contrast	0 ~ 2 (Can be set only in easy day type custom mode.) ex> easyDayContrast=2	
easyDayColors	Easy setting Daytime Colors	0 ~ 2 (Can be set only in easy day type custom mode.) ex> easyDayColors=1	
easyDayBrightness	Easy setting Daytime Brightnes	0 ~ 2 (Can be set only in easy day type custom mode.) ex> easyDayBrightness=2	
easyNightType	Easy setting Nighttime mode	0 ~ 3 ex> easyNightType=0 0 : Custom 1 : Natural 2 : Vivid 3 : Denoise	
easyNightSharpness	Easy setting Nighttime sharpness	0 ~ 3 (Can be set only in easy night type custom mode.) ex> easyNightSharpness=3	
easyNightGamma	Easy setting Nighttime contrast	0 ~ 2 (Can be set only in easy night type custom mode.) ex> easyNightGamma=2	
easyNightBrightness	Easy setting Nighttime brightness	0~2 (Can be set only in easy night type custom mode.) ex> easyNightBrightness=1	

Receive Parameters

Parameters	**Description**	**Values**	**ETC**
returnCode	Return code	Refer Return Code	

Example

Send data:

easyDayType=0&easyDaySharpness=3&easyDayContrast=0&easyDayColors=1&easyDayBrightness=2&easyNightType=1&action=videoEasySetting&mode=0

Response data:

returnCode=0

16. Video – Image

※ It does not support autoAdjust, sharpeness, noise filter, 3dnr setting when self adjust video mode is supported.

16.1. Read

Get video-image information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoImage	
mode	[Mode Code](#)	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer [Return Code](#)	
mirroring	Image mirroring	off | horizontal | vertical | both ex> mirroring=vertical	
pivot	Image pivot	off | clockwise | counterclockwise ex> pivot=off	
defog	Defog	on | off ex> defog=off	Group5
colorStyle	Color style	pccolor | tvcolor ex> colorStyle=pccolor	Group1,3,4
backLightCompensation	Back light compensation	off | on | hsblc ex> backLightCompensation=off	
autoAdjust	Auto adjustment	on | off ex> autoAdjust=on	Group5
sharpness	Image sharpness	0 ~ 6 ex> sharpness=3	Group1,2,3,4
		0 ~ 6, auto	Group5
noiseFilter	Noise filter strength	0 ~ 6, auto ex> noiseFilter=3	Group1,5
		0 ~ 15	Group2
		0 ~ 6	Group3,4
3dnrFilter	3DNR filter strength	0~2 ex>3dnrFilter=3	Group1,3,4
		0~15	Group2
		0~7	Group5
sensorMode	Sensor mode	4:3 | 16:9 ex> sensorMode=16:9	Sensor Mode setting support model
sensorFrameRate	Sensor frame rate	60 | 50 | 30 | 25 ex> sensorFrameRate=30	

Example

Send data:

action=videoImage&mode=1

Response data:

returnCode=0&mirroring=off&pivot=off&colorStyle=pccolor&backLightCompensation=off&sharpness=3&noiseFilter=3&3dnrFilter=1&sensorFrameRate=30

16.2. Write

Set video-image information.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

<table>
<tr><th>Parameters</th><th>Description</th><th>Values</th><th>ETC</th></tr>
<tr><td>action</td><td>Action ID</td><td>videoImage</td><td></td></tr>
<tr><td>mode</td><td>Mode Code</td><td>0</td><td></td></tr>
<tr><td>mirroring</td><td>Image mirroring</td><td>off | horizontal | vertical | both
ex> mirroring=vertical</td><td></td></tr>
<tr><td>pivot</td><td>Image pivot</td><td>off | clockwise | counterclockwise
ex> pivot=off</td><td></td></tr>
<tr><td>defog</td><td>Defog</td><td>on | off
ex> defog=off</td><td>Group5</td></tr>
<tr><td>colorStyle</td><td>Color style</td><td>pccolor | tvcolor
ex> colorStyle=pccolor</td><td>Group1,3,4</td></tr>
<tr><td>backLightCompensation</td><td>Back light compensation</td><td>off | on | hsblc
ex> backLightCompensation=off</td><td></td></tr>
<tr><td>autoAdjust</td><td>Auto adjustment</td><td>on | off
ex> autoAdjust=on</td><td>Group5</td></tr>
<tr><td rowspan="2">sharpness</td><td rowspan="2">Image sharpness</td><td>0 ~ 6
ex> sharpness=3</td><td>Group1,2,3,4</td></tr>
<tr><td>0 ~ 6, auto</td><td>Group5</td></tr>
<tr><td rowspan="3">noiseFilter</td><td rowspan="3">Noise filter strength</td><td>0 ~ 6, auto
ex> noiseFilter=3</td><td>Group1,5</td></tr>
<tr><td>0 ~ 15</td><td>Group2</td></tr>
<tr><td>0 ~ 6</td><td>Group3,4</td></tr>
<tr><td rowspan="3">3dnrFilter</td><td rowspan="3">3DNR filter strength</td><td>0~2
ex>3dnrFilter=3</td><td>Group1,3,4</td></tr>
<tr><td>0~15</td><td>Group2</td></tr>
<tr><td>0~7</td><td>Group5</td></tr>
<tr><td>sensorMode</td><td>Sensor mode</td><td>4:3 | 16:9
ex> sensorMode=16:9</td><td>Sensor Mode setting support model</td></tr>
<tr><td>sensorFrameRate</td><td>Sensor frame rate</td><td>60 | 50 | 30 | 25
ex> sensorFrameRate=30</td><td></td></tr>
</table>

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

```
action=videoImage&mode=0&mirroring=off&pivot=off&colorStyle=pccolor&backLightCompensation=off&sharpness=3&noiseFilter=3
&3dnrFilter=1&sensorFrameRate=30
```

Response data:

```
returnCode=0
```

17. Video – White Balance

17.1. Read

Get white balance controls.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoWb	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
wbMode	White balance preset	auto | incandescent | 4000k | 5000k | sunny | cloudy | flash | fluorescent_warm | fluorescent_cold | manual | hold ex> mode=auto	Group1
		auto | auto_wide | auto_full | 2500k | 3200k | 4200k | 5800k | 9500k| color_temperature ex> mode=auto	Group2
redGain	red gain when use manual mode	10 ~ 500 ex> redGain=168	Group1
blueGain	blue gain when use manual mode	10 ~ 500 ex> blueGain=100	Group1
colorTemperature	color temperature when use color temperature WB mode	0~63 ex> colorTemperature=31	Group2

Example

Send data:
action=videoWb&mode=1

Response data:
returnCode=0&wbMode =auto&redGain=168&blueGain=152

17.2. **Write**

Set white balance controls.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	**Description**	**Values**	**ETC**
action	Action ID	videoWb	
mode	Mode Code	0	
wbMode	White balance preset	auto | incandescent | 4000k | 5000k | sunny | cloudy | flash | fluorescent_warm | fluorescent_cold | manual | hold ex> mode=auto	Group1
		auto | auto_wide | auto_full | 2500k | 3200k | 4200k | 5800k | 9500k| color_temperature ex> mode=auto	Group2
redGain	red gain when use manual mode	10 ~ 500 ex> redGain=168	Group1
blueGain	blue gain when use manual mode	10 ~ 500 ex> blueGain=100	Group1
colorTemperature	color temperature when use color temperature WB mode	0~63 ex> colorTemperature=31	Group2

Receive Parameters

Parameters	**Description**	**Values**	**ETC**
returnCode	Return code	Refer Return Code	

Example

Send data:
action=videoWb&mode=0&wbMode=auto&redGain=168&blueGain=100

Response data:
returnCode=0

18. Video – Exposure

18.1. **Read**

Get video exposure controls.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoExposure	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
targetGain	Auto exposure Target gain	-10 ~ 10 ex> targetGain=0	
wdr	WDR or Digital WDR	off | auto | x2 | x3 | x4 ex> ~~localExposure=off~~ ▶ wdr=off	Group1
		on | off ex> wdr=off	Gruop2
antiFlicker	Anti-flicker mode (Hz) (disable when manual ae control on)	off | indoor50 | indoor60 | outdoor50 | outdoor60 ex> antiFlicker=off	
slowShutter	Max slow shutter speed (disable when manual ae control on)	off | 1/15s | 1/7.5s ex> slowShutter=1/15s	
irisControlMode	enable dc iris control	auto (group1 & group2 camera only) fullopen | dcIrisAuto (with DC-IRIS) manual | pIrisAuto (with P-IRIS) off ex> irisControlMode=auto	
minIris	minimum iris value with P-IRIS manual mode	0~100 ex> minIris=0	
maxIris	maximum iris value with P-IRIS manual mode	0~100 ex> maxIris=100	
irisPosition	Iris position with P-IRIS manual mode	1~ irisMaxPosition	
manualAeControl	Use manual AE control	on | off ex> manualAeControl =off	
lowerShutterLimit	Lower limit of shutter speed (1/n sec)	30 ~ 10000 (Group5) 30 ~ 8000 (Group1,2,3,4) ex> lowerShutterLimit=30	
upperShutterLimit	Upper limit shutter speed (1/n sec)	30 ~ 10000 (Group5) 30 ~ 8000 (Group1,2,3,4) ex> upperShutterLimit=10000	

lowerGainLimit	Lower limit of sensor gain when manual ae control on	1dB | 2dB | 3dB | … 40dB | 41dB | 42dB | … ex> lowerGainLimit=3dB	
upperGainLimit	Upper limit of sensor gain when manual ae control on	1dB | 2dB | 3dB | … 40dB | 41dB | 42dB | … ex> upperGainLimit=3dB	
wdrLevel	WDR Level	1 | 2 | 3 ex> wdrLevel=2	Activated only when WDR is 'on'.

Example

Send data:

action=videoExposure&mode=1

Response data:

returnCode=0&targetGain=0&wdr=on&antiFlicker=off&slowShutter=off&manualAeControl=on&lowerShutterLimit=30&upperShutterLimit=30&lowerGainLimit=3dB&upperGainLimit=3dB&irisControlMode=fullopen&wdrLevel=2

18.2. **Write**

Set video exposure controls.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoExposure	
mode	Mode Code	0	
targetGain	Target gain	-10 ~ 10 ex> targetGain=0	
wdr	WDR or Digital WDR	off | auto | x2 | x3 | x4 ex> ~~localExposure=off~~ ▶ wdr=off	Group1
		on | off ex> wdr=off	Gruop2
wdrLevel	WDR Level	1 | 2 | 3 ex> wdrLevel=2	Activated only when WDR is ‘on’.
antiFlicker	Anti-flicker mode (Hz) (disable when manual ae control on)	off | indoor50 | indoor60 | outdoor50 | outdoor60 ex> antiFlicker=off	
slowShutter	Max slow shutter speed (disable when manual ae control on)	off | 1/15s | 1/7.5s ex> slowShutter=1/15s	
irisControlMode	enable dc iris control	auto (group1 & group2 camera only) fullopen | dcIrisAuto (with DC-IRIS) manual | pIrisAuto (with P-IRIS) off ex> irisControlMode=auto	
irisPosition	Iris position with P-IRIS manual mode	1~ irisMaxPosition	
manualAeControl	Use manual AE control	on | off ex> manualAeControl =off	
lowerShutterLimit	Lower limit of shutter speed (1/n sec)	30 ~ 10000 (Group5) 30 ~ 8000 (Group1,2,3,4) ex> lowerShutterLimit=30	
upperShutterLimit	Upper limit shutter speed (1/n sec)	30 ~ 10000 (Group5) 30 ~ 8000 (Group1,2,3,4) ex> upperShutterLimit=10000	
lowerGainLimit	Lower limit of sensor gain when manual ae control on	iso100 | iso200 | iso300 | iso400 | iso600 | iso800 | iso1600 | iso3200 | iso6400 | iso12800 | iso25600 ex> lowerGainLimit=iso100	legacy
		1dB | 2dB | 3dB | … 40dB | 41dB | 42dB … | maxGain ex> lowerGainLimit=3dB	
upperGainLimit	Upper limit of sensor gain when manual ae control on	iso100 | iso200 | iso300 | iso400 | iso600 | iso800 | iso1600 | iso3200 | iso6400 | iso12800 | iso25600 ex> upperGainLimit=iso6400	legacy
		1dB | 2dB | 3dB | … 40dB | 41dB | 42dB … | maxGain	

		ex> upperGainLimit=3dB	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=videoExposure&mode=0&targetGain=0&localExposure=off&wdr=on&wdrLevel=2&antiFlicker=off&slowShutter=off&manualAeControl=on&lowerShutterLimit=30&upperShutterLimit=30&lowerGainLimit=3dB&upperGainLimit=3dB&irisControlMode=fullopen

Response data:
returnCode=0

19. Video – Day&Night

19.1. Read

Get day and night mode controls.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoDaynight	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
bwMode	Black and white mode	on | off | auto | schedule ex> bwMode=auto	
icrMode	IR cur filter mode	on | off | auto | schedule ex> icrMode=auto	
refocus	refocus when dan & night switch	on | off | dnmodeshift | iradptiveshift ex> refocus=off	
dnSwitchLevel	brightness leven when switch night mode from day mode	1 ~ 10 ex> dnSwitchBrightness=3	
		1 ~ 8 ex> dnSwitchBrightness=3	Group5
ndSwitchLevel	brightness leven when switch day mode from night mode	1 ~ 10 ex> ndSwitchBrightness=3	
		1 ~ 8 ex> ndSwitchBrightness=3	Group5
schedule	Day & Night schedule	Refer Schedule Information	

Example

Send data:

action=videoDaynight&mode=1

Response data:

returnCode=0&bwMode=auto&icrMode=auto&refocus=0ff&dnSwitchBrightness=3&ndSwitchBrightness=3&schedule=0000000000000000155555
5555555555540000000000000000_000000000000000155555555555555540000000000000000_000000000000000155555555555555540000000
00000000_000000000000000155555555555555540000000000000000_000000000000000155555555555555540000000000000000_000000000
000000155555555555555540000000000000000_000000000000000155555555555555540000000000000000

19.2. **Write**

Set day and night mode controls.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoDaynight	
mode	Mode Code	0	
bwMode	Black and white mode	on | off | auto | schedule ex> bwMode=auto	
icrMode	IR cur filter mode	on | off | auto | schedule ex> icrMode=auto	
refocus	refocus when dan & night switc	on | off | dnmodeshift | iradptiveshift ex> refocus=off	
dnSwitchLevel	brightness leven when switch night mode from day mode	1 ~ 10 ex> dnSwitchLevel=3	
		1 ~ 8 ex> dnSwitchLevel=3	Group5
ndSwitchLevel	brightness leven when switch day mode from night mode	1 ~ 10 ex> ndSwitchLevel=3	
		1 ~ 8 ex> ndSwitchLevel=3	Group5
schedule	Day & Night schedule	Refer Schedule Information	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=videoDaynight&mode=0&bwMode=auto&icrMode=auto&refocus=off&dnSwitchBrightness=3&ndSwitchBrightness=3&schedule=0000000
0000000001555555555555555400000000000000_0000000000000000155555555555555540000000000000000_0000000000000000155555555
55555554000000000000000_00000000000000001555555555555555400000000000000000_000000000000000015555555555555554000000000
00000_000000000000000015555555555555554000000000000000_000000000000000015555555555555554000000000000000

Response data:

returnCode=0&

20. Video – Miscellaneous

20.1. Read

Get miscellaneous video controls.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoMisc	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
videoOutType	CVBS video output	ntsc | pal ex> videoOutType=ntsc	
irStrength	IR brightness control (%)	auto | 100 | 80 | 60 | 40 | 20 | 0 ex> irStrength=auto	
imageStabilizer	get image stabilizer	on | off ex> imageStabilizer=off	
smartIR	get smartIR	on | off | auto ex> smartIR=off	smartIR support model

Example

Send data:
action=videoMisc&mode=1

Response data:
returnCode=0&videoOutType=ntsc&irStrength =auto&smartIR=auto

20.2. **Write**

Set miscellaneous video controls.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoMisc	
mode	Mode Code	0	
videoOutType	CVBS video output	ntsc | pal ex> videoOutType=ntsc	
irStrength	IR brightness control (%)	auto | 100 | 80 | 60 | 40 | 20 | 0 ex> irStrength=auto	
imageStabilizer	set image stabilizer on	on | off ex> imageStabilizer=off	
smartIR	set smartIR	on | off | auto ex> smartIR=off	smartIR support model

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=videoMisc&mode=0&videoOutType=ntsc&irStrength =auto&smartIR=on

Response data:
returnCode=0

21. Video – Streaming (except model group 4)

21.1. Read

Get video streaming information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoStreaming	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
intelligentCodec	Use intelligent codec	on | off ex> intelligentCodec=on	intelligent Codec support model
		off | low | normal | high ex> intelligentCodec=high	intelligent Codec level support model
useStream{N}	Use N-th stream	on | off ex> useStream1=on	
codecStream{N}	Stream encoding CODEC	h264 | h265 | jpeg ex> codecStream1=h264	
resolutionStream{N}	Stream resolution	3840x2160 | 2304x1296 | 1920x1080 | 1280x720 | 704x480 | 640x360 | 352x240 2592x1944 | 2048x1536 | 1600x1200 | 1280x960 | 800x600 | 640x480 | 480x360 | 320x240 ex> resolutionStream1=1920x1080	
qualityStream{N}	Stream quality	basic | standard | high | veryHigh or 512~10240 ex> qualityStream1=veryHigh	
bitrateControlStream{N}	Bitrate control mode	vbr | cbr ex> bitrateControlStream1=cbr	
framerateStream{N}	Stream frame per second	30 | 15 | 10 | 5 | 4 | 3 | 2 | 1 ex> framerateStream1=30	
{N} : 1 ~ 3(4)			
useMultiview{M}		on | off ex>	
width{M}	Multiview screen width	352 ~ width ex>	
height{M}	Multiview screen height	240 ~ resolution Height	
left{M}	Multiview screen left position		
top{M}	multi-view screen top position		

{M} : 2 ~ 3(4)

Example

Send data:
action=videoStreaming&mode=1

Response data:
returnCode=0&intelligentCodec=on&useStream1=on&codecStream1=h264&resolutionStream1=1920x1080&qualityStream1=veryHigh&bitrateControlStream1=cbr&framerateStream1=30&useStream2=on&codecStream2=h264&resolutionStream2=1280x720&qualityStream2=veryHigh&bitrateControlStream2=cbr&framerateStream2=5&useStream3=on&codecStream3=h264&resolutionStream3=352x240&qualityStream3=veryHigh&bitrateControlStream3=cbr&framerateStream3=5

21.2. Write

Set video streaming information.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoStreaming	
mode	Mode Code	0	
intelligentCodec	Use intelligent codec	on | off ex> intelligentCodec=on	
useStream{N}	Use N-th stream	on | off ex> useStream1=on	
codecStream{N}	Stream encoding CODEC	h264 | h265 | jpeg ex> codecStream1=h264	
resolutionStream{N}	Stream resolution	2304x1296 | 1920x1080 | 1280x720 | 704x480 | 352x240 ex> resolutionStream1=1920x1080	
qualityStream{N}	Stream quality	basic | standard | high | veryHigh | 512~10240Kbps ex> qualityStream1=veryHigh	
bitrateControlStream{N}	Bitrate control mode	vbr | cbr ex> bitrateControlStream1=cbr	
framerateStream{N}	Stream frame per second	30 | 15 | 10 | 5 | 4 | 3 | 2 | 1 ex> framerateStream1=30	
{N} : 1 ~ 3(4)			
useMultiview{M}		on | off ex>	
width{M}	Multiview screen width	352 ~ width ex>	
height{M}	Multiview screen height	240 ~ resolution Height	
left{M}	Multiview screen left position		
top{M}	multi-view screen top position		
{M} : 2 ~ 3(4)			

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=videoStreaming&mode=0&intelligentCodec=on&useStream1=on&codecStream1=h264&resolutionStream1=1920x1080&qualityStream1=veryHigh&bitrateControlStream1=cbr&framerateStream1=30&useStream2=on&codecStream2=h264&resolutionStream2=1280x720&qualityStream2=veryHigh&bitrateControlStream2=cbr&framerateStream2=5&useStream3=on&codecStream3=h264&resolutionStream3=352x240&qualityStream3=veryHigh&bitrateControlStream3=cbr&framerateStream3=5

Response data:

returnCode=0

22. Video – MAT

22.1. Read

Get video MAT(Motion Adaptive Transmission) control.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoMat	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useMat	Use MAT on/off	on | off ex> useMat=on	
sensitivity	MAT sensitivity	1 ~ 5 ex> sensitivity=3	
inactivityPeriod	Detection threshold for inactivity (seconds)	1 ~ 60 ex> inactivityPeriod=10	
framerateStream{N}	Frame rate of N-th stream at inactivity	off | 15 | 10 | 5 | 4 | 3 | 2 | 1 ex> framerateStream1=5	
{N} : 1 ~ 3			

Example

Send data:
action=videoMat&mode=1

Response data:
returnCode=0&useMat=on&sensitivity=3&inactivityPeriod=10&framerateStream1=5&framerateStream2=5&framerateStream3=off

22.2. Write

Set video MAT(Motion Adaptive Transmission) control.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoMat	
mode	Mode Code	0	
useMat	Use MAT on/off	on | off ex> useMat=on	
sensitivity	MAT sensitivity	1 ~ 5 ex> sensitivity=3	
inactivityPeriod	Detection threshold for inactivity (seconds)	1 ~ 60 ex> inactivityPeriod=10	
framerateStream{N}	Frame rate of N-th stream at inactivity	off | 15 | 10 | 5 | 4 | 3 | 2 | 1 ex> framerateStream1=5	
{N} : 1 ~ 3			

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=videoMat&mode=0&useMat=on&sensitivity=3&inactivityPeriod=10&framerateStream1=5&framerateStream2=5&framerateStream3=off

Response data:
returnCode=0

23. Video – Privacy Mask

23.1. Read

Get privacy zone mask.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoPrivacy	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
usePrivacy	Use privacy mask on/off	on | off ex>usePrivacy=on	
maxWidth	Max zone width	<integer> ex> maxWidth=80	
maxHeight	Max zone height	<integer> ex> maxWidth=45	
useZone{N}	Use N-th zone on/off	on | off ex>useZone1=on	
nameZone{N}	Name of N-th zone	<string> ex>nameZone1=sample	
topZone{N}	Y position of top-left	0 ~ <max height> ex> topZone1=10	
leftZone{N}	X position of top-left	0 ~ <max width> ex> leftZone1=10	
bottomZone{N}	Y position of bottom-right	0 ~ <max height > ex> bottomZone1=100	
rightZone{N}	X position of bottom-right	0 ~ <max width> ex> rightZone1=100	
{N} : 1 ~ 8			Group5
{N} : 1 ~ 16			Group1,2 ,3,4

Example

Send data:
action=videoPrivacy&mode=1

Response data:
returnCode=0&usePrivacy=on&maxWidth=80&maxHeight=45&useZone1=on&nameZone1=&leftZone1=16&topZone1=0&rightZone1=48&bottomZone1=31&useZone2=on&nameZone2=&leftZone2=16&topZone2=0&rightZone2=48&bottomZone2=31&useZone3=off&useZone4=off&useZone5=off&useZone6=off&useZone7=off&useZone8=off&useZone9=off&useZone10=off&useZone11=off&useZone12=off&useZone13=off&useZone14=off&useZone15=off&useZone16=off

23.2. **Write**

Set privacy zone mask.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoPrivacy	
mode	Mode Code	0	
usePrivacy	Use privacy mask on/off	on | off ex>usePrivacy=on	
useZone{N}	Use N-th zone on/off	on | off ex>useZone1=on	
nameZone{N}	Name of N-th zone	<string> ex>nameZone1=sample	
topZone{N}	Y position of top-left	0 ~ <bottomZone> ex> topZone1=10	
leftZone{N}	X position of top-left	0 ~ <rightZone> ex> leftZone1=10	
bottomZone{N}	Y position of bottom-right	<topZone> ~ <max height > ex> bottomZone1=100	
rightZone{N}	X position of bottom-right	<leftZone> ~ <max width> ex> rightZone1=100	
{N} : 1 ~ 8			Group5
{N} : 1 ~ 16			Group1,2,3,4

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=videoPrivacy&mode=0&usePrivacy=on&useZone1=on&nameZone1=privacy1&topZone1=10&leftZone1=20&bottomZone1=40&rightZone1=50&useZone2=off&nameZone2=&topZone2=10&leftZone2=20&bottomZone2=40&rightZone2=50&useZone3=off&useZone4=off&useZone5=off&useZone6=off&useZone7=off&useZone8=off&useZone9=off&useZone10=off&useZone11=off&useZone12=off&useZone13=off&useZone14=off&useZone15=off&useZone16=off

Response data:

returnCode=0

24. Video – Snapshot (group 5,6)

24.1. Read

Get jpg snapshot image

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoSnapshot	
mode	Mode Code	1	
streamIndex	Set the stream index to transm	Normal : 1 FishEye : 1,2 ex> streamIndex=1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=videoSnapshot&mode=1&streamIndex=1

25. Video stream snapshot

Get video snapshot

※ The desired stream must be set to jpeg.

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoStreamSnapshot	
mode	Mode Code	1	
streamIndex	Stream index for snapshot	1~ number of streams supported by the camera. ex) streamIndex=4	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Receive image

Header	Description	Values	ETC
response	Response message	HTTP/1.1 200 OK	
Content-type	Content type	image/webp	
Data	JPEG image	Stream data	

Example

Send data:

action=videoStreamSnapshot&mode=1&streamIndex=4

Response data:

{{JPEG binary data}}

26. Video – OSD Text

26.1. Read

Get video Text OSD information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoOsdText	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useOsd	Use text OSD	on | off ex> useOsd=on	
text	OSD string	<string> ex> text=TextMessage	
textSize	OSD font size	0 | 1 | 2 | 3 | 4 Ex> textSize=4	'0' is smallest size, '4' is largest size
textColor	OSD font Color	<string> ex> textColor=ffffff	
textTransparency	OSD font transparency	255 | 200 | 145 | 95 | 40 ex> textTransparency=255	The smaller the number, the more transparent it is.
positionX	OSD x poisiton	0~100 ex> positionX=0	
positionY	OSD y poisiton	0~100 ex> positionY=0	

Example

Send data:
action=videoOsdText&mode=1

Response data:
returnCode=0&useOsd=on&text=TextMessage&textSize=4&textColor=ffffff&textTransparency=255&positionX=50&positionY=50

26.2. **Write**

Set video OSD information.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useOsd	Use text OSD	on | off ex> useOsd=on	
text	OSD string	<string> ex> text=TextMessage	
textSize	OSD font size	0 | 1 | 2 | 3 | 4 Ex> textSize=4	'0' is smallest size, '4' is largest size
textColor	OSD font Color	<string> ex> textColor=ffffff	
textTransparency	OSD font transparency	255 | 200 | 145 | 95 | 40 ex> textTransparency=255	The smaller the number, the more transparent it is.
positionX	OSD x poisiton	0~100 ex> positionX=0	
positionY	OSD y poisiton	0~100 ex> positionY=0	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=videoOsdText&mode=0&useOsd=on&text=TextMessage&textSize=4&textColor=ffffff&textTransparency=255&positionX=50&positionY=50
Response data:
returnCode=0

27. Video – OSD DateTime

27.1. Read

Get video DateTime OSD information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	videoOsdDateTime	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useOsd	Use text OSD	on | off ex> useOsd=on	
dateFormat	Date format	YYYY/MM/DD | MM/DD/YYYY | DD/MM/YYYY ex> dateFormat= YYYY/MM/DD	
timeForamt	Time format	HH:MM:SS | II:MM:SS_PP | PP_II:MM:SS ex> timeFormat=HH:MM:SS	
textSize	OSD font size	0 | 1 | 2 | 3 | 4 Ex> textSize=4	'0' is smallest size, '4' is largest size
textColor	OSD font Color	<string> ex> textColor=ffffff	
textTransparency	OSD font transparency	255 | 200 | 145 | 95 | 40 ex> textTransparency=255	The smaller the number, the more transparent it is.
positionX	OSD x poisiton	0~100 ex> positionX=0	
positionY	OSD y poisiton	0~100 ex> positionY=0	

Example

Send data:
action=videoOsdDateTime&mode=1

Response data:
returnCode=0&useOsd=on&dateFormat=YYYY/MM/DD&timeFormat=HH:MM:SS&textSize=0&textColor=ffffff&textTransparency=255&positionX=50&positionY=50

27.2. **Write**

Set video DateTime OSD information.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useOsd	Use text OSD	on | off ex> useOsd=on	
dateFormat	Date format	YYYY/MM/DD | MM/DD/YYYY | DD/MM/YYYY ex> dateFormat= YYYY/MM/DD	
timeForamt	Time format	HH:MM:SS | II:MM:SS_PP | PP_II:MM:SS ex> timeFormat=HH:MM:SS	
textSize	OSD font size	0 | 1 | 2 | 3 | 4 Ex> textSize=4	‘0’ is smallest size, ‘4’ is largest size
textColor	OSD font Color	<string> ex> textColor=ffffff	
textTransparency	OSD font transparency	255 | 200 | 145 | 95 | 40 ex> textTransparency=255	The smaller the number, the more transparent it is.
positionX	OSD x poisiton	0~100 ex> positionX=0	
positionY	OSD y poisiton	0~100 ex> positionY=0	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=videoOsdDateTime&mode=0&useOsd=on&dateFormat=YYYY/MM/DD&timeFormat=HH:MM:SS&textSize=0&textColor=ffffff&textTransparency=255&positionX=50&positionY=50&
Response data:
returnCode=0

28. Http MJPEG streaming

Get 25. Http MJPEG stream

※ The desired stream must be set to jpeg.

Syntax

http[s]://<device ip>:<port>/video/mjpeg?<parameter>=<value>

Send Parameters

Parameters	Description	Values	ETC
trackID	Steram ID	1~ number of streams supported by the camera. ex) trackID=1	The codec for that stream must be set to jpeg.

Receive data

Header	Description	Values	ETC
response	Response message	HTTP/1.1 200 OK	
Content-type	Content type	Multipart/x-mixed-replace	
Data	HTTP MJPEG stream	MJPEG Stream data	

Example

Send data:

http://<device ip>/video/mjpeg?trackID=1

Response data:

{{HTTP MJPEG stream}}

29. Audio

29.1. Read

Get audio input/output controls.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	audio	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
audioCodec	Audio encoding CODEC	g711_ulaw | adpcm | g726 ex> audioCodec=g726	Group1
		g711_ulaw | g711_alaw| g726 ex> audioCodec=g726	Group2
useInput	Use Input on/off	on | off ex> useInput=on	
inputType	select input audio type	lineIn | microphone ex> inputType=lineIn	
inputVolume	Input volume	0 ~ 16 ex> inputVolume=8	
useOutput	Use Output on/off	on | off ex> useOutput=on	
outputVolume	Output volume	0 ~ 16 ex> outputVolume=8	
noiseFilter	Use noise filter	on | off ex> noiseFilter=on	Noise filter support model

Example

Send data:

action=audio&mode=1

Response data:

returnCode=0&audioCodec=g726&useInput=on&inputVolume=8&useOutput=off&outputVolume=8

29.2. **Write**

Set audio input/output controls.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	audio	
mode	Mode Code	0	
audioCodec	Audio encoding CODEC	g711_ulaw | adpcm | g726 ex> audioCodec=g726	Group1
		g711_ulaw | g711_alaw| g726 ex> audioCodec=g726	Group2
useInput	Use Input on/off	on | off ex> useInput=on	
inputType	select input audio type	lineIn | microphone ex> inputType=lineIn	
inputVolume	Input volume	0 ~ 16 ex> inputVolume=8	
useOutput	Use Output on/off	on | off ex> useOutput=on	
outputVolume	Output volume	0 ~ 16 ex> outputVolume=8	
noiseFilter	Use noise filter	on | off ex> noiseFilter=on	Noise filter support model

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=audio&mode=0&audioCodec=g726&useInput=on&inputVolume=8&useOutput=off&outputVolume=8

Response data:

returnCode=0

30. Event Action – Alarm out

30.1. Read

Get alarm output controls for event action.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionAlarmout	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useAlarmOut	Use alarm output on/off	on | off ex> useAlarmOut=on	
useAlarmOut{N}	Use N-th alarm output on/off	on | off ex> useAlarmOut1=on	Multiple alarm out supports model
dwellTime	Dwell of alarm output (seconds)	5 | 10 | 15 | 20 | 25 | 30 | 40 | 50 | 60 | 75 | 90 | 105 | 120 | 180 | 240 | 300 | 360 | 420 | 480 | 540 | 600 | 900 ex> dwellTime=20	
dwellTime{N}	Dwell of N-th alarm output (seconds)	5 | 10 | 15 | 20 | 25 | 30 | 40 | 50 | 60 | 75 | 90 | 105 | 120 | 180 | 240 | 300 | 360 | 420 | 480 | 540 | 600 | 900 ex> dwellTime1=20	Multiple alarm out supports model
scheduleStart	Start time of alarm output schedule (per 15 minute)	hh:mm ex> scheduleStart=00:00	
scheduleStart{N}	Start time of N-th alarm output schedule (per 15 minute)	hh:mm ex> scheduleStart2=00:00	Multiple alarm out supports model
scheduleEnd	End time of alarm output schedule (per 15 minute)	hh:mm ex> scheduleEnd=24:00	
scheduleEnd{N}	End time of N-th alarm output schedule (per 15 minute)	hh:mm ex> scheduleEnd2=24:00	Multiple alarm out supports model

Example

Send data:
action=actionAlarmout&mode=1

Response data:
returnCode=0&useAlarmOut=on&dwellTime=20&scheduleStart=00:00&scheduleEnd=24:00

returnCode=0&useAlarmOut1=off&dwellTime1=15&scheduleStart1=00:15&scheduleEnd1=24:00&useAlarmOut2=on&dwellTime2=25&scheduleStart2=03:00&scheduleEnd2=23:15

30.2. **Write**

Set alarm output controls for event action.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionAlarmout	
mode	Mode Code	0	
useAlarmOut	Use alarm output on/off	on | off ex> useAlarmOut=on	
useAlarmOut{N}	Use N-th alarm output on/off	on | off ex> useAlarmOut1=on	multiple alarm out supports model
dwellTime	Dwell of alarm output (seconds)	5 | 10 | 15 | 20 | 25 | 30 | 40 | 50 | 60 | 75 | 90 | 105 | 120 | 180 | 240 | 300 | 360 | 420 | 480 | 540 | 600 | 900 ex> dwellTime=20	
dwellTime{N}	Dwell of N-th alarm output (seconds)	5 | 10 | 15 | 20 | 25 | 30 | 40 | 50 | 60 | 75 | 90 | 105 | 120 | 180 | 240 | 300 | 360 | 420 | 480 | 540 | 600 | 900 ex> dwellTime1=20	multiple alarm out supports model
scheduleStart	Start time of alarm output schedule (per 15 minute)	hh:mm ex> scheduleStart=00:00	
scheduleStart{N}	Start time of N-th alarm output schedule (per 15 minute)	hh:mm ex> scheduleStart2=00:00	multiple alarm out supports model
scheduleEnd	End time of alarm output schedule (per 15 minute)	hh:mm ex> scheduleEnd=24:00	
scheduleEnd{N}	End time of N-th alarm output schedule (per 15 minute)	hh:mm ex> scheduleEnd2=24:00	multiple alarm out supports model

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=actionAlarmout&mode=0&useAlarmOut=on&dwellTime=20&scheduleStart=00:00&scheduleEnd=24:00

action=actionAlarmout&mode=0&useAlarmOut1=off&dwellTime1=15&scheduleStart1=00:15&scheduleEnd1=24:00&useAlarmOut2=on&dwellTime2=25&scheduleStart2=03:00&scheduleEnd2=23:15

Response data:

returnCode=0

31. Event Action – Email

31.1. Read

Get email information for event action.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionEmail	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useEmail	Use email on/off	on | off ex> useEmail=on	
smtpServer	SMTP server address	<string> ex> smtpServer=mail.sample.com	
smtpPort	SMTP server port	<integer> ex> smtpPort=25	
useSSLTLS	Encrypted Connection Type	on | off ex> useSSLTLS=off	Group1,2
		on | off | ssl The value on is the startTLS ex> useSSLTLS=on	Group3,4,5
id	SMTP login ID	<string> ex> id=mailuser	
sender	Sender information	<string> ex> sender=camera1	
recipientList	Recipient list	<string> [| <string>] ex> recipientList=usr1@sample.com|usr2@sample.com	

Example

Send data:

action=actionEmail&mode=1

Response data:

returnCode=0&useEmail=on&smtpServer=mail.sample.com&smtpPort=25&useSSLTLS=off&id=mailuser&sense=camera&recipentList=usr1@sample.com|usr2@other.com

31.2. **Write (Secured)**

Set email information for event action. This command can be only used with https or RSA encryption.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionEmail	
mode	Mode Code	0 | 5 ex> mode=0	
useEmail	Use email on/off	on | off ex> useEmail=on	
smtpServer	SMTP server address	<string> ex> smtpServer=mail.sample.com	
smtpPort	SMTP server port	<integer> ex> smtpPort=25	
useSSLTLS	Encrypted Connection Type	on | off ex> useSSLTLS=off	Group1,2
		on | off | ssl The value on is the startTLS ex> useSSLTLS=on	Group3,4,5
id	SMTP login ID	<string> ex> id=mailuser	
password	SMTP login password	If the mode is 0, then this command can be only used with https. <string> ex> password=passwd	
		If mode is 5 < hexadecimal string encypted By RSA public Key> ex> password=30a8dcc18f9f7dad53381ff7bfe561b0f0fc00f9d2d5b5dbb4066ae448369321ad1d7d3b2ea157bca8e8787b9ec79c81d07fa8230ff992668a6e262fc1e4261d86a15d3b449760371ff0376a42b351d8203dae4c5f204ec8a48b06e34ff9ed98ecf2c7c192d380277214be4e9f3df8293cf29d6ab0938a96248f907c965fb25e	
sender	Sender information	<string> ex> sender=camera1	
recipientList	Recipient list	<string> [| <string>] ex>recipientList=usr1@sample.com|usr2@sample.com	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=actionEmail&mode=0&useEmail=on&smtpServer=mail.sample.com&smtpPort=25&useSSLTLS=off&id=mailuser&password=passwd&sender=camera&recipientList=usr1@sample.com|usr2@other.com

Response data:
returnCode=0

31.3. **Test SMTP server (Secured)**

Set email information for event action. This command can be only used with https or RSA encryption.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	**Description**	**Values**	**ETC**
action	Action ID	actionEmail	
mode	Mode Code	2 | 7 ex> mode=2	
smtpServer	SMTP server address	<string> ex> smtpServer=mail.sample.com	
smtpPort	SMTP server port	<integer> ex> smtpPort=25	
useSSLTLS	Encrypted Connection Type	on | off ex> useSSLTLS=off	Group1,2
		on | off | ssl The value on is the startTLS ex> useSSLTLS=ssl	Group 3~6
id	SMTP login ID	<string> ex> id=mailuser	
password	SMTP login password	If the mode is 2, then this command can be only used with https. <string> ex> password=passwd	
		If mode is 7 < hexadecimal string encypted By RSA public Key> ex> password=30a8dcc18f9f7dad53381ff7bfe561b0f0fc00f9d2d5b5dbb4066ae448369321ad1d7d3b2ea157bca8e8787b9ec79c81d07fa8230ff992668a6e262fc1e4261d86a15d3b449760371ff0376a42b351d8203dae4c5f204ec8a48b06e34ff9ed98ecf2c7c192d380277214be4e9f3df8293cf29d6ab0938a96248f907c965fb25e	
sender	Sender information	<string> ex> sender=camera1	
recipientList	Recipient list	<string> [| <string>] ex>recipientList=usr1@sample.com|usr2@sample.com	

Receive Parameters

Parameters	**Description**	**Values**	**ETC**
returnCode	Return code	Refer Return Code	

Example

Send data:

action=actionEmail&mode=2&smtpServer=mail.sample.com&smtpPort=25&useSSLTLS=off&id=mailuser&password=passwd&sender=camera1&recipientList=usr1@sample.com|usr2@other.com

Response data:

returnCode=0

32. Event Action – Remote Callback

32.1. Read

Get remote callback information for event action.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionCallback	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useCallback	Use remote callback on/off	on | off ex> useCallback=on	
retryCount	Callback retry count at failure	1 ~ 10 ex> retryCount=1	
remoteSite{N}	Callback server address	<IPv4 address> ex> remoteSite1=192.168.1.140	
portSite{N}	Callback server port	<integer> 8000 ~ 12000 ex> portSite1=8201	
{N} : 1 ~ 5			

Example

Send data:

action=actionCallback&mode=1

Response data:

returnCode=0&useCallback=on&retryCount=3&remoteSite1=10.0.126.70&portSite1=8201&remoteSite2=10.0.130.10&portSite2=830
1

32.2. **Write**

Set remote callback information for event action.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionCallback	
mode	Mode Code	0	
useCallback	Use remote callback on/off	on | off ex> useCallback=on	
retryCount	Callback retry count at failure	1 ~ 10 ex> retryCount=1	
remoteSite{N}	Callback server address	<IPv4 address> ex> remoteSite1=192.168.1.140	
portSite{N}	Callback server port	<integer> 8000 ~ 12000 ex> portSite1=8201	
{N} : 1 ~ 5			

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=actionCallback&mode=0&useCallback=on&retryCount=3&remoteSite1=10.0.126.70&portSite1=8201&remoteSite2=10.0.130.10&portSite2=8301

Response data:

returnCode=0

33. Event Action – Audio Alarm

33.1. Read

Get audio alarm information for event action.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionAudioAlarm	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useAudioAlarm	Use audio alarm on/off	on | off ex> useAudioAlarm=on	
audioFileURL	URL for audio file	<string> ex> audioFileURL=	
audioFileCount	number of audio alarm file	<integer> ex> audioFileCount=1	
audioFileName{N}	audio alarm file name	<String> ex> audioFileName1=hello.wav	
{N} : 1 ~ audioFileCount			

Example

Send data:
action= actionAudioAlarm&mode=1

Response data:
returnCode=0&useAudioAlarm=on&audioFileCount=1&audioFileName1=hello.wav

33.2. **Write**

Set audio alarm information information for event action.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionAudioAlarm	
mode	Mode Code	0	
useAudioAlarm	Use audio alarm on/off	on | off ex> useAudioAlarm=on	
removeFileName	audio file name for remove	<string> ex> removeFileName=hello.wav	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=actionAudioAlarm&mode=0&useAudioAlarm=off&removeFileName=hello.wav

Response data:
returnCode=0

33.3. Upload file

upload audio alarm file for event action.

Make sure that preparing for File Upload is preceded. Please refer Preparing File Upload.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionAudioAlarm	
mode	Mode Code	9001	
fileName	audio file name for upload	<string> ex> fileName=hello.wav	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=actionAudioAlarm&mode=9001&filename=hello.wav

Response data:

returnCode=0

34. Event Action – FTP

34.1. Read

Get FTP information for event action.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionFtp	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useFTP	Use FTP on/off	on | off ex> useFTP=on	
ftpServer{N}	FTP server address	<IPv4 address> ex> ftpServer1=10.0.126.70	
uploadPath{N}	FTP Upload path	<string> ex> uploadPath1=eventImage	
port{N}	FTP server port	<integer> ex>portSite1=21	
userID{N}	FTP user ID	<string> ex>userID1=user	
uploadType	Upload type	event | always ex> uploadType=event	
uploadFrequency	Upload frequency	when upload type is alwasys 1s | 1m | 5m | 10m | 15m | 20m | 30m | 1h | 5h | 10h | 15h | 20h | 30h when upload type is event 1s | 30m | 20m | 15m | 12m ex> uploadFrequency=1m	
duration	Upload duration when upload type is event	5sec | 10sec | 15sec | 30sec | overall ex> duration=5s	
resolution	Upload image resolution	704x480 | 640x512 | 640x360 | 352x240 ex> resolution=352x240	
quality	Upload image quality	basic | standard | high | veryHigh ex> quality=veryHigh	
prefix	Upload file prefix	<string> ex>prefix=image	
namingType	Upload file naming type	datetime | sequence | overwrite ex> namingType=datetime	
maxNumberSequence	Max number of file sequence	1 ~ 1000000 ex>maxNumberOfNamingSequence=1000	
{N}: 1 ~ 2			

Example

Send data:

action=actionFtp&mode=1

Response data:

returnCode=0&useFtp=on&ftpServer1=10.0.126.70&uploadPath1=event&port1=21&userID1=user&uploadType=event&uploadFrequency=1m&resolution=352x240&quality=basic&prefix=image&namingType=datetime

34.2. Write (Secured)

Set FTP information for event action. This command can be only used with https or RSA encryption.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionFtp	
mode	Mode Code	0 | 5 ex> mode=0	
useFTP	Use FTP on/off	on | off ex> useFTP=on	
ftpServer{N}	FTP server address	<IPv4 address> ex> ftpServer1=10.0.126.70	
uploadPath{N}	FTP Upload path	<string> ex> uploadPath1=eventImage	
port{N}	FTP server port	<integer> ex>portSite1=21	
userID{N}	FTP user ID	<string> ex>userID1=user	
password{N}	FTP user password	If the mode is 0, then this command can be only used with https. <string> ex>passwrd1=passwd	
		If mode is 5 < hexadecimal string encypted By RSA public Key> ex> password1=30a8dcc18f9f7dad53381ff7bfe561b0f0fc00f9d2d5b5dbb4066ae448369321ad1d7d3b2ea157bca8e8787b9ec79c81d07fa8230ff992668a6e262fc1e4261d86a15d3b449760371ff0376a42b351d8203dae4c5f204ec8a48b06e34ff9ed98ecf2c7c192d380277214be4e9f3df8293cf29d6ab0938a96248f907c965fb25e	
uploadType	Upload type	event | always ex> uploadType=event	
uploadFrequency	Upload frequency	when upload type is alwasys 1s | 1m | 5m | 10m | 15m | 20m | 30m | 1h | 5h | 10h | 15h | 20h | 30h when upload type is event 1s | 30m | 20m | 15m | 12m ex> uploadFrequency=1m	
duration	Upload duration when upload type is event	5sec | 10sec | 15sec | 30sec | overall ex> duration=5s	
resolution	Upload image resolution	704x480 | 640x512 | 640x360 | 352x240 ex> resolution=352x240	
quality	Upload image quality	basic | standard | high | veryHigh ex> quality=veryHigh	

prefix	Upload file prefix	<string> ex>prefix=image	
namingType	Upload file naming type	datetime | sequence | overwrite ex> namingType=datetime	
maxNumberOfNamingSequence	Max number of file sequence	1 ~ 10000 ex>maxNumberOfNamingSequence=1000	
{N}: 1 ~ 2			

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=actionFtp&mode=0&useFtp=on&ftpServer1=10.0.126.70&uploadPath1=event&port1=21&userID1=user&password1=passwd&uploadType=event&uploadFrequency=1m&resolution=352x240&quality=basic&prefix=image&namingType=datetime

Response data:

returnCode=0

34.3. **Test FTP server (Secured)**

FTP upload test. This command can be only used with https or RSA encryption.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionFtp	
mode	Mode Code	2 | 7 ex> mode=2	
ftpServer	FTP server address	<IPv4 address> ex> ftpServer=10.0.126.70	
uploadPath	FTP Upload path	<string> ex> uploadPath=eventImage	
port	FTP server port	<integer> ex>portSite=21	
userID	FTP user ID	<string> ex>userID=user	
password	FTP user password	If the mode is 2, then this command can be only used with https. <string> ex>passwrd=passwd	
		If mode is 7 < hexadecimal string encypted By RSA public Key> ex> password=30a8dcc18f9f7dad53381ff7bfe561b0f0fc00f9d2d5b5dbb4066ae448369321ad1d7d3b2ea157bca8e8787b9ec79c81d07fa8230ff992668a6e262fc1e4261d86a15d3b449760371ff0376a42b351d8203dae4c5f204ec8a48b06e34ff9ed98ecf2c7c192d380277214be4e9f3df8293cf29d6ab0938a96248f907c965fb25e	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code return 0 when success , return 500 when upload failed	

Example

Send data:

action=actionFtp&mode=0&ftpServer=10.0.126.70&uploadPath=event&port=21&userID=user&password=passwd

Response data:

returnCode=0

35. Event Action – SD Recording

35.1. Read

Get SD card recording controls for event action.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionRecord	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useRecord	use SD recording on/off	on | off ex> useRecord=on	
recordAudio	recording with audio stream	on | off ex> recordAudio=on	
scheduleMode	SD record schedule mode	alwaysevent | alwaystimelapse | alwayseventtimelapse | schedule ex> scheduleMode=schedule	
preEventDuration	duration for pre-event recording (sec)	0 | 5 | 10 | 15 | 20 | 25 | 30 | 40 | 50 | 60 ex> preEventDuration=10	
postEventDuration	duration for post-event recording (sec)	0 | 5 | 10 | 15 | 20 | 25 | 30 | 40 | 50 | 60 ex> postEventDuration=0	
schedule	SD record schedule	Refer Schedule Information 0: none 1: event 2: time-lapse 3: time-lapse/event	
timelapseRecordingStream	set video stream when recording time-lapse	primary | secondary | tertiary | quinary | sextus ex> timelapseRecordingStream=secondary	
eventRecordingStream	set video stream when recording event	primary | secondary | tertiary | quinary | sextus ex> eventRecordingStream=primary	
recordingPreference	priority for recording stream setting	none | sd | network ex> recordingPreference=none	
networkRecordingFailover	use SD record when network connection failure	off | always | onschedule ex> networkRecordingFailover=off	

Example

Send data:
action=actionRecord&mode=1

Response data:
returnCode=0&&useRecord=on&recordAudio=on&scheduleMode=schedule&preEventDuration=10&postEventDuration=0&schedule=&timelapseRecordingStream=secondary&eventRecordingStream=primary&recordingPreference=none&networkRecordingFailover=off

35.2. **Write**

Set SD card recording controls for event action.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	actionRecord	
mode	Mode Code	0	
useRecord	use SD recording on/off	on | off ex> useRecord=on	
recordAudio	recording with audio stream	on | off ex> recordAudio=on	
scheduleMode	SD record schedule mode	alwaysevent | alwaystimelapse | alwayseventtimelapse | schedule ex> scheduleMode=schedule	
preEventDuration	duration for pre-event recording (sec)	0 | 5 | 10 | 15 | 20 | 25 | 30 | 40 | 50 | 60 ex> preEventDuration=10	
postEventDuration	duration for post-event recording (sec)	0 | 5 | 10 | 15 | 20 | 25 | 30 | 40 | 50 | 60 ex> postEventDuration=0	
schedule	SD record schedule	Refer Schedule Information 0: none 1: event 2: time-lapse 3: time-lapse/event	
timelapseRecordingStream	set video stream when recording time-lapse	primary | secondary | tertiary | quinary | sextus ex> timelapseRecordingStream=secondary	
eventRecordingStream	set video stream when recording event	primary | secondary | tertiary | quinary | sextus ex> eventRecordingStream=primary	
recordingPreference	priority for recording stream setting	none | sd | network ex> recordingPreference=none	
networkRecordingFailover	use SD record when networ connection failure	off | always | onschedule ex> networkRecordingFailover=off	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=actionRecord&mode=0&useRecord=on&recordAudio=on&scheduleMode=schedule&preEventDuration=10&postEventDuration=0&schedule=&timelapseRecordingStream=secondary&eventRecordingStream=primary&recordingPreference=none&networkRecordingFailover=off

Response data:

returnCode=0

36. Event – Alarm In

36.1. Read

Get alarm input information for event.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventAlarmin	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useAlarmIn	Use alarm input on/off	on | off ex> useAlarmIn=on	
useAlarmIn{N}	Use N-th alarm input on/off	on | off ex> useAlarmIn1=on	Multiple alarm in supports model
alarmName	Name of alarm input	<string> ex> alarmName=AlarmIn	
alarmName{N}	Name of N-th alarm input	on | off ex> alarmName1=AlarmIn	Multiple alarm in supports model
alarmType	Alarm input type	no | nc ex> alarmType=no	
alarmType{N}	N-th alarm input type	on | off ex> alarmType2=no	Multiple alarm in supports model
referenceEvent	Reference event	none | motionDetection | tripZone ex> referenceEvent=tripZone referenceEvent=tripZone|motionDetection	Reference event supports model
referenceEvent{N}	N-th reference event	none | motionDetection | tripZone ex> referenceEvent1=tripZone referenceEvent2=tripZone|motionDetection	Multiple alarm in supports model
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut{N}	Use N-th alarm output action on/off	on | off ex> actionAlarmOut2=on	Multiple alarm in supports model
actionAlarmOut{N}	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut2=1|2	Multiple alarm in / multiple alarm out
actionDayNight	Use D/N action on/off	on | off	

		ex> actionDayNight=off	
actionDayNight1	Use 1th D/N action on/off (D/N setting is available only in alarmIn1.)	on | off ex> actionDayNight1=off	Multiple alarm in supports model
actionBlackWhite	Use B/W action on/off	on | off ex> actionBlackWhite=off	
actionBlackWhite1	Use 1th B/W action on/off (B/W setting is available only in alarmIn1.)	on | off ex> actionBlackWhite1=off	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmail{N}	Use N-th email action on/off	on | off ex> actionEmail1=on	Multiple alarm in supports model
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionEmailAttachImage{N}	Use N-th email image attach action on/off	on | off ex> actionEmailAttachImage2=off	Multiple alarm in supports model
actionCallback	Use remote callback action on/off	on | off actionCallback=on	
actionCallback{N}	Use N-th remote callback action on/off	on | off actionCallback1=on	Multiple alarm in supports model
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionCallbackSite{N}	Use N-th remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite2=1|3	Multiple alarm in supports model
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
actionAudioAlarm{N}	Use N-th audio alarm action on/off	on | off ex> actionAudioAlarm2=off	Multiple alarm in supports model
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
audioAlarmName{N}	N-th audio alarm file	<string> ex> actionAudioAlarm1=audio.wav	Multiple alarm in supports model
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionFTPupload{N}	Use N-th FTP action on/off	on | off ex> actionFTPupload2=off	Multiple alarm in supports model
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
actionRecord{N}	Use N-th SD card recording action on/off	on | off ex> actionRecord2=off	Multiple alarm in supports model

actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	
actionMovePTZTo{N}	Use move to N-th PTZ preset position action on/off	on | off ex> actionMovePTZTo1=off	Multiple alarm in supports model
presetID	preset positon for actionMovePTZTo	1~16 ex> presetID=1	
presetID{N}	N-th preset positon for actionMovePTZTo	1~16 ex> presetID2=1	Multiple alarm in supports model

Example

Send data:
action=eventAlarmin&mode=1

Response data:
returnCode=0&useAlarmIn=off&alarmName=Alarm In&alarmType=no&actionAlarmOut=on&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

returnCode=0&useAlarmIn1=on&alarmName1=Alarm In 1&alarmType1=no&actionAlarmOut1=2&actionDayNight1=off&actionEmail1=off&actionCallback1=off&actionAudioAlarm1=off&actionFTPupload1=off&actionRecord1=off&actionMovePTZTo1=off&useAlarmIn2=on&alarmName2=Alarm In 2&alarmType2=no&actionAlarmOut2=1&actionEmail2=off&actionCallback2=off&actionAudioAlarm2=off&actionFTPupload2=off&actionRecord2=off&actionMovePTZTo2=off&useAlarmIn3=off&alarmName3=Alarm In 3&alarmType3=no&actionAlarmOut3=&actionEmail3=off&actionCallback3=off&actionAudioAlarm3=off&actionFTPupload3=off&actionRecord3=off&actionMovePTZTo3=off&useAlarmIn4=on&alarmName4=Alarm In 4&alarmType4=no&actionAlarmOut4=1|2&actionEmail4=off&actionCallback4=off&actionAudioAlarm4=off&actionFTPupload4=off&actionRecord4=off&actionMovePTZTo4=off

36.2. Write

Set alarm input information for event.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventAlarmin	
mode	[Mode Code]()	0	
useAlarmIn	Use alarm input on/off	on | off ex> useAlarmIn=on	
useAlarmIn{N}	Use N-th alarm input on/off	on | off ex> useAlarmIn1=on	Multiple alarm in supports model
alarmName	Name of alarm input	<string> ex> alarmName=AlarmIn	
alarmName{N}	Name of N-th alarm input	on | off ex> alarmName1=AlarmIn	Multiple alarm in supports model
alarmType	Alarm input type	no | nc ex> alarmType=no	
alarmType{N}	N-th alarm input type	on | off ex> alarmType2=no	Multiple alarm in supports model
referenceEvent	Reference event	none | motionDetection | tripZone ex> referenceEvent=tripZone referenceEvent=tripZone|motionDetection	Reference event supports model
referenceEvent{N}	N-th reference event	none | motionDetection | tripZone ex> referenceEvent1=tripZone referenceEvent2=tripZone|motionDetection	Multiple alarm in supports model
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut{N}	Use N-th alarm output action on/off	on | off ex> actionAlarmOut2=on	Multiple alarm in supports model
actionAlarmOut{N}	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut2=1|2	Multiple alarm in / multiple alarm out
actionDayNight	Use D/N action on/off	on | off ex> actionDayNight=off	
actionDayNight1	Use 1th D/N action on/off (D/N setting is available only in alarmIn1.)	on | off ex> actionDayNight1=off	Multiple alarm in supports model

actionBlackWhite	Use B/W action on/off	on | off ex> actionBlackWhite=off	
actionBlackWhite1	Use 1th B/W action on/off (B/W setting is available only in alarmIn1.)	on | off ex> actionBlackWhite1=off	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmail{N}	Use N-th email action on/off	on | off ex> actionEmail1=on	Multiple alarm in supports model
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=eventAlarmin&mode=0&useAlarmIn=off&alarmName=AlarmIn&alarmType=no&actionAlarmOut=on&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

action=eventAlarmin&mode=0&useAlarmIn1=on&alarmName1=AlarmIn1&alarmType1=no&actionAlarmOut1=2&actionDayNight1=off&actionEmail1=off&actionCallback1=off&actionAudioAlarm1=off&actionFTPupload1=off&actionRecord1=off&actionMovePTZTo1=off&useAlarmIn2=on&alarmName2=AlarmIn2&alarmType2=no&actionAlarmOut2=1&actionEmail2=off&actionCallback2=off&actionAudioAlarm2=off&actionFTPupload2=off&actionRecord2=off&actionMovePTZTo2=off&useAlarmIn3=off&alarmName3=Alarm In 3&alarmType3=no&actionAlarmOut3=&actionEmail3=off&actionCallback3=off&actionAudioAlarm3=off&actionFTPupload3=off&actionRecord3=off&actionMovePTZTo3=off&useAlarmIn4=on&alarmName4=AlarmIn4&alarmType4=no&actionAlarmOut4=1|2&actionEmail4=off&actionCallback4=off&actionAudioAlarm4=off&actionFTPupload4=off&actionRecord4=off&actionMovePTZTo4=off

Response data:

returnCode=0

37. Event – PIR Detection

37.1. Read

Get PIR detection information for event.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventPIR	PIR support model
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
usePIR	Use alarm input on/off	on | off ex> usePIR=on	
sensitivity	PIR sensitivity	1 ~ 5 ex> sensitivity=3	
useIgnoringTime	Use ignoring time on/off	on | off ex> useIgnoringTime=on	
ignoringTimeStart	Ignoring time start (per 15 minutes)	hh:mm ex> ignoringTimeStart=07:00	
ignoringTimeEnd	Ignoring time end (per 15 minutes)	hh:mm ex> ignoringTimeEnd=18:00	
ignoreInterval	Detection ignore interval (seconds)	0 ~ 5 ex> ignoreInterval=2	
referenceEvent	Reference event	none | motionDetection | tripZone ex> referenceEvent=tripZone referenceEvent=tripZone|motionDetection	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	

audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	

Example

Send data:
action=eventPIR&mode=1

Response data:
returnCode=0&referenceEvent=motionDetection&actionAlarmOut=on&actionEmail=on&actionEmailAttachImage=off&actionCallback=off&actionAudioAlarm=off&actionFTPupload=off&actionRecord=on&usePIR=on&sensitivity=3&useIgnoringTime=off&ignoringTimeStart=18:15&ignoringTimeEnd=10:00&ignoreInterval=1

37.2. **Write**

Set PIR detection information for event.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventPIR	
mode	Mode Code	0	
returnCode	Return code	Refer Return Code	
usePIR	Use alarm input on/off	on | off ex> usePIR=on	
sensitivity	PIR sensitivity	1 ~ 5 ex> sensitivity=3	
useIgnoringTime	Use ignoring time on/off	on | off ex> useIgnoringTime=on	
ignoringTimeStart	Ignoring time start (per 15 minutes)	hh:mm ex> ignoringTimeStart=07:00	
ignoringTimeEnd	Ignoring time end (per 15 minutes)	hh:mm ex> ignoringTimeEnd=18:00	
ignoreInterval	Detection ignore interval (seconds)	0 ~ 5 ex> ignoreInterval=2	
referenceEvent	Reference event	none | motionDetection | tripZone ex> referenceEvent=tripZone referenceEvent=tripZone|motionDetection	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

usePIR=on&sensitivity=3&useIgnoringTime=off&ignoringTimeStart=18:15&ignoringTimeEnd=10:00&ignoreInterval=1&referenceEvent=motionDetection&actionAlarmOut=on&actionEmail=on&actionEmailAttachImage=off&actionCallback=off&actionAudioAlarm=off&actionFTPupload=off&actionRecord=on&action=eventPIR&mode=0

Response data:

returnCode=0

38. Event – Motion Detection

38.1. Read

Get motion detection information for event.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventMotion	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useMotion	Use motion detection on/off	on | off ex> useMotion=on	
daySensitivity	Motion sensitivity for daytime	1 ~ 5 ex> daySensitivity=3	
dayMinBlock	Min motion blocks for daytime	1 ~ 330 ex> dayMinBlock=10	
nightSensitivity	Motion sensitivity for nighttime	1 ~ 5 ex> nightSensitivity=3	
nightMinBlock	Min motion blocks for nighttime	1 ~ 330 ex> nightMinBlock=10	
detectionOnPTZMovement	Set enable detection on PTZ movement	on | off ex> detectionOnPTZMovement=off	
motionZone	Motion zone area	Refer Motion Block Information	
ignoreInterval	Motion ignore interval (seconds)	0 ~ 5 ex> ignoreInterval=3	
daytimeStart	Daytime start (per 15 minutes)	hh:mm ex> daytimeStart=07:00	
daytimeEnd	Daytime end (per 15 minutes)	hh:mm ex> daytimeEnd=18:00	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	

actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	
presetID	proset positon for actionMovePTZTo	1~16 ex> presetID=1	

Example

Send data:
action=eventMotion&mode=1

Response data:
returnCode=0&useMotion=on&daySensitiviti=3&dayMinBlock=10&nightSensitivity=3&nightMinBlock=10&montionZone=&ignoreInterval=2&daytimeStart=&daytimeEnd=&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

38.2. **Write**

Set motion detection information for event.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventMotion	
mode	Mode Code	0	
useMotion	Use motion detection on/off	on | off ex> useMotion=on	
daySensitivity	Motion sensitivity for daytime	1 ~ 5 ex> daySensitivity=3	
dayMinBlock	Min motion blocks for daytime	1 ~ 330 ex> dayMinBlock=10	
nightSensitivity	Motion sensitivity for nighttime	1 ~ 5 ex> nightSensitivity=3	
nightMinBlock	Min motion blocks for nighttime	1 ~ 330 ex> nightMinBlock=10	
detectionOnPTZMovement	Set enable detection on PTZ movement	on | off ex> detectionOnPTZMovement=off	
motionZone	Motion zone area	Refer Motion Block Information	
ignoreInterval	Motion ignore interval (seconds)	0 ~ 5 ex> ignoreInterval=3	
daytimeStart	Daytime start (per 15 minutes)	hh:mm ex> daytimeStart=07:00	
daytimeEnd	Daytime end (per 15 minutes)	hh:mm ex> daytimeEnd=18:00	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	

actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	
presetID	proset positon for actionMovePTZTo	1~16 ex> presetID=1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=eventMotion&mode=0&useMotion=on&daySensitiviti=3&dayMinBlock=10&nightSensitivity=3&nightMinBlock=10&montionZone=&ignoreInterval=2&daytimeStart=&daytimeEnd=&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

Response data:

returnCode=0

39. Event – Trip Zone

39.1. Read

Get trip zone information for event.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventTripzone	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useTripzone	Use trip zone on/off	on | off ex> useTripzone=on	
daySensitivity	Trip zone sensitivity for daytime	1 ~ 5 ex> daySensitivity=3	
nightSensitivity	Trip zone sensitivity for nighttime	1~ 5 ex> nightSensitivity=3	
tripZone	Trip zone area	Refer Motion Block Information	
detectionType	Trip zone detection type	in | out | both ex> detectionType=both	
ignoreInterval	Motion ignore interval (seconds)	0 ~ 5 ex> ignoreInterval=3	
daytimeStart	Daytime start (per 15 minutes)	hh:mm ex> daytimeStart=07:00	
daytimeEnd	Daytime end (per 15 minutes)	hh:mm ex> daytimeEnd=18:00	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string>	

		ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	
presetID	proset positon for actionMovePTZTo	1~16 ex> presetID=1	

Example

Send data:
action=eventTripzone&mode=1

Response data:
returnCode=0&useTripzone=on&daySensitiviti=3&nightSensitivity=3&inZone=&outZone=&ignoreInterval=2&daytimeStart=&daytimeEnd=&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

39.2. **Write**

Set trip zone information for event.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventTripzone	
mode	Mode Code	0	
useTripzone	Use trip zone on/off	on | off ex> useTripzone=on	
daySensitivity	Trip zone sensitivity for daytime	1 ~ 5 ex> daySensitivity=3	
nightSensitivity	Trip zone sensitivity for nighttime	1~ 5 ex> nightSensitivity=3	
tripZone	Trip zone area	Refer Motion Block Information	
detectionType	Trip zone detection type	in | out | both ex> detectionType=both	
ignoreInterval	Motion ignore interval (seconds)	0 ~ 5 ex> ignoreInterval=3	
daytimeStart	Daytime start (per 15 minutes)	hh:mm ex> daytimeStart=07:00	
daytimeEnd	Daytime end (per 15 minutes)	hh:mm ex> daytimeEnd=18:00	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	

actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	
presetID	proset positon for actionMovePTZTo	1~16 ex> presetID=1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=eventTripzone&mode=0&useTripzone=on&daySensitivity=3&nightSensitivity=3&tripZone=&detectionType=both&ignoreInterval=2&daytimeStart=&daytimeEnd=&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

Response data:
returnCode=0

40. Event – Audio Detection

40.1. Read

Get audio detection information for event.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventAudiodetect	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useAudioDetection	Use audio detection on/off	on | off ex> useAudioDetection=on	
sensitivity	Audio detection sensitivity	1 ~ 5 ex> sensitivity=3	
triggerTime	Detection trigger time (x 100ms)	0 ~ 30 ex> triggerTime=10	
usingDeactiveTime	Using detect deactivation time on/off	on | off ex> usingDeactiveTime=off	
deactiveTimeStart	Detect deactivation time start (per 15 minutes)	hh:mm ex> deactiveTimeStart=07:00	
deactiveTimeEnd	Detect deactivation time start (per 15 minutes)	hh:mm ex> deactiveTimeEnd=18:00	
ignoreInterval	Detection ignore interval (seconds)	0 ~ 5 ex> ignoreInterval=2	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	

actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	
presetID	proset positon for actionMovePTZTo	1~16 ex> presetID=1	

Example

Send data:
action=eventAudiodetect&mode=1

Response data:
returnCode=0&useAudioDetection=on&sensitivity=3&triggerTime=10&usingDeactiveTime=off&deactiveTimeStart=07:00&deactiveTimeEnd=18:00&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

40.2. **Write**

Set audio detection information for event.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventAudiodetect	
mode	[Mode Code]()	0	
useAudioDetection	Use audio detection on/off	on | off ex> useAudioDetection=on	
sensitivity	Audio detection sensitivity	1 ~ 5 ex> sensitivity=3	
triggerTime	Detection trigger time (x 100ms)	0 ~ 30 ex> triggerTime=10	
usingDeactiveTime	Using detect deactivation time on/off	on | off ex> usingDeactiveTime=off	
deactiveTimeStart	Detect deactivation time start (per 15 minutes)	hh:mm ex> deactiveTimeStart=07:00	
deactiveTimeEnd	Detect deactivation time start (per 15 minutes)	hh:mm ex> deactiveTimeEnd=18:00	
ignoreInterval	Detection ignore interval (seconds)	0 ~ 5 ex> ignoreInterval=2	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	

actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	
presetID	proset positon for actionMovePTZTo	1~16 ex> presetID=1	

Receive Parameters

Parameters	**Description**	**Values**	**ETC**
returnCode	Return code	Refer Return Code	

Example

Send data:
action=eventAudiodetect&mode=0&useAudioDetection=on&sensitivity=3&triggerTime=10&usingDeactiveTime=off&deactiveTimeStart=07:00&deactiveTimeEnd=18:00&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

Response data:
returnCode=0

41. Event – Tampering

41.1. Read

Get tampering detection information for event.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventTampering	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useTampering	Use tampering detection on/off	on | off ex> useTampering=on	
sensitivity	Tampering detection sensitivity	1 ~ 5 ex> sensitivity=3	
triggerTime	Detection trigger time (x 100ms)	0 ~ 30 ex> triggerTime=10	
usingDeactiveTime	Using detect deactivation time on/off	on | off ex> usingDeactiveTime=off	
deactiveTimeStart	Detect deactivation time start (per 15 minutes)	hh:mm ex> deactiveTimeStart=07:00	
deactiveTimeEnd	Detect deactivation time start (per 15 minutes)	hh:mm ex> deactiveTimeEnd=18:00	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	

actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	
presetID	proset positon for actionMovePTZTo	1~16 ex> presetID=1	

Example

Send data:
action=eventTampering&mode=1

Response data:
returnCode=0&useTampering=on&sensitivity=3&triggerTime=10&usingDeactiveTime=off&deactiveTimeStart=07:00&deactiveTimeEnd=18:00&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

41.2. **Write**

Set tampering detection information for event.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	**Description**	**Values**	**ETC**
action	Action ID	eventTampering	
mode	[Mode Code]()	0	
useTampering	Use tampering detection on/off	on | off ex> useTampering=on	
sensitivity	Tampering detection sensitivity	1 ~ 5 ex> sensitivity=3	
triggerTime	Detection trigger time (x 100ms)	0 ~ 30 ex> triggerTime=10	
usingDeactiveTime	Using detect deactivation time on/off	on | off ex> usingDeactiveTime=off	
deactiveTimeStart	Detect deactivation time start (per 15 minutes)	hh:mm ex> deactiveTimeStart=07:00	
deactiveTimeEnd	Detect deactivation time start (per 15 minutes)	hh:mm ex> deactiveTimeEnd=18:00	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	

actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	
presetID	proset positon for actionMovePTZTo	1~16 ex> presetID=1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=eventTarmpering&mode=0&useTampering=on&sensitivity=3&triggerTime=10&usingDeactiveTime=off&deactiveTimeStart=07:00&deactiveTimeEnd=18:00&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

Response data:
returnCode=0

42. Event – Auto Tracking

42.1. Read

Get auto tracking information for event.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventAutoTracking	Auto tracking support model
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useAutoTracking	Use auto tracking on/off	on | off ex> useAutoTracking=on	
usingZoom	Use zoom for auto tracking	on | off ex> usingZoom =3	
objectSize	Set the size of the object to be tracked with auto tracking.	Small | Medium | Large ex> objectSize=Large	
waitTimeAfterTrace	Set wait time after trace	0 | 30 | 60 | 90 | 120 | 150 | 180 | 210 | 240 | 270 | 300 ex> waitTimeAfterTrace=90	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	

Example

Send data:

action= eventAutoTracking&mode=1

Response data:

returnCode=0&actionAlarmOut=on&actionEmail=on&actionEmailAttachImage=on&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=off&actionFTPupload=on&actionRecord=on&useAutoTracking=on&usingZoom=on&objectSize=Large&waitTimeAfterTrace=90

42.2. **Write**

Set tampering detection information for event.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventTampering	
mode	Mode Code	0	
useAutoTracking	Use auto tracking on/off	on | off ex> useAutoTracking=on	
usingZoom	Use zoom for auto tracking	on | off ex> usingZoom =3	
objectSize	Set the size of the object to be tracked with auto tracking.	Small | Medium | Large ex> objectSize=Large	
waitTimeAfterTrace	Set wait time after trace	0 | 30 | 60 | 90 | 120 | 150 | 180 | 210 | 240 | 270 | 300 ex> waitTimeAfterTrace=90	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionAlarmOut	Use N-th alarm output action on/off	1 | 2 ex> actionAlarmOut=1|2	Multiple alarm in supports model
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
useAutoTracking	Use auto tracking on/off	on | off ex> useAutoTracking=on	
usingZoom	Use zoom for auto tracking	on | off ex> usingZoom =3	
objectSize	Set the size of the object to be tracked with auto tracking.	Small | Medium | Large ex> objectSize=Large	

waitTimeAfterTrace	Set wait time after trace	0 | 30 | 60 | 90 | 120 | 150 | 180 | 210 | 240 | 270 | 300 ex> waitTimeAfterTrace=90	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=eventAutoTracking&mode=0&useAutoTracking=on&usingZoom=on&objectSize=Large&waitTimeAfterTrace=90&actionAlarmOut=on&actionEmail=on&actionEmailAttachImage=on&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=off&actionFTPupload=on&actionRecord=on

Response data:

returnCode=0

43. Event – Face Detection

43.1. Read

Get face detection information for event.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventFaceDetection	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useFaceDetection	Use face detection on/off	on | off ex> useFaceDetection=on	
sensitivity	face detection sensitivity	1 ~ 5 ex> sensitivity=3	
minFaceSize	Min face size for face detection event	1 ~ 5 ex> minFaceSize=5	
maxDetectCount	max face detection area count	8 | 32 ex> maxDetectCount=8	
ignoreInterval	Motion ignore interval (seconds)	0 ~ 5 ex> ignoreInterval=3	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	

presetID	proset positon for actionMovePTZTo	1~16 ex> presetID=1	

Example

Send data:

action=eventFaceDetection&mode=1

Response data:

returnCode=0&useFaceDetection=on&sensitivity=3&minFaceSize=3&maxDetectCount=8&ignoreInterval=2&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

43.2. **Write**

Set motion detection information for event.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventMotion	
mode	Mode Code	0	
useFace	Use face detection on/off	on | off ex> useFace=on	
sensitivity	face detection sensitivity	1 ~ 5 ex> sensitivity=3	
minFaceSize	Min face size for face detection event	1 ~ 5 ex> minFaceSize=5	
maxDetectCount	max face detection area count	8 | 32 ex> maxDetectCount	
ignoreInterval	Motion ignore interval (seconds)	0 ~ 5 ex> ignoreInterval=3	
actionAlarmOut	Use alarm output action on/off	on | off ex> actionAlarmOut=on	
actionEmail	Use email action on/off	on | off ex> actionEmail=on	
actionEmailAttachImage	Use email image attach action on/off	on | off ex> actionEmailAttachImage=off	
actionCallback	Use remote callback action on/off	on | off ex> actionCallback=on	
actionCallbackSite	Use remote callback sites	1 | 2 | 3 | 4 | 5 ex> actionCallbackSite=1|3	
actionAudioAlarm	Use audio alarm action on/off	on | off ex> actionAudioAlarm=off	
audioAlarmName	Audio alarm file	<string> ex> actionAudioAlarm=audio.wav	
actionFTPupload	Use FTP action on/off	on | off ex> actionFTPupload=off	
actionRecord	Use SD card recording action on/off	on | off ex> actionRecord=off	
actionMovePTZTo	Use move to PTZ preset position action on/off	on | off ex> actionMovePTZTo=off	
presetID	proset positon for actionMovePTZTo	1~16 ex> presetID=1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=eventFaceDetection&mode=0&useFaceDetection=on&sensitivity=3&minFaceSize=3&maxDetectCount=8&ignoreInterval=2&actionEmail=on&actionEmailAttachImage=off&actionCallback=on&actionCallbackSite=1|2&actionAudioAlarm=no&actionFTPupload=off&actionRecord=off

Response data:

returnCode=0

44. Event – System Event

44.1. Read

Get system event information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventSystem	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
systemAliveCheck	Use system alive check on/off	on | off ex> useTempering =on	
systemAliveInterval	System alive check interval (hours)	1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 18 | 24 | 168 | 360 | 720 ex> systemAliveInterval =1	
systemAliveEmail	Use email action for system alive check	on | off ex> systemAliveEmail =on	
systemAliveCallback	Use remote callback action for system alive check	on | off ex> systemAliveCallback =on	
systemAliveCallbackSite	Use remote callback sites for system alive check	1 | 2 | 3 | 4 | 5 ex> systemAliveCallbackSite =1|3	
alarmInCheck	Use alarm input check on/off	on | off ex> alarmInCheck =on	
alrmInCheckInterval	Alarm input check interval (hours)	1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 18 | 24 | 168 | 360 | 720 ex> alrmInCheckInterval =1	
alarmInCheckEmail	Use email action for alarm input check	on | off ex> alarmInCheckEmail=off	
alarmInCheckCallback	Use remote callback action for alarm input check	on | off ex> alarmInCheckCallback =on	
alarmInCheckCallbackSite	Use remote callback sites for alarm input check	1 | 2 | 3 | 4 | 5 ex> alarmInCheckCallbackSite =1|3	
sdCheck	Use SD card check on/off	on | off ex> sdCheck =on	
sdCheckkEmail	Use email action SD card check	on | off ex> sdCheckEmail=off	
sdCheckCallback	Use remote callback action for SD card check	on | off ex> sdCheckCallback =on	
sdCheckCallbackSite	Use remote callback sites for SD card check	1 | 2 | 3 | 4 | 5 ex> sdCheckCallbackSite =1|3	

Example

Send data:

action=eventSystem&mode=1

Response data:

returnCode=0&systemAliveCheck=on&systemAliveInterval=1&systemAliveEmail=on&systemAliveCallback=on&systemAliveCallbackSite=1|2&alarmInCheck=on&alarmInCheckInterval=2&alarmInCheckEmail=off&alarmInCheckCallback=off&sdCheck=off

44.2. **Write**

Set system event information.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventSystem	
mode	Mode Code	0	
systemAliveCheck	Use system alive check on/off	on | off ex> useTempering =on	
systemAliveInterval	System alive check interval (hours)	1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 18 | 24 | 168 | 360 | 720 ex> systemAliveInterval =1	
systemAliveEmail	Use email action for system alive check	on | off ex> systemAliveEmail =on	
systemAliveCallback	Use remote callback action for system alive check	on | off ex> systemAliveCallback =on	
systemAliveCallbackSite	Use remote callback sites for system alive check	1 | 2 | 3 | 4 | 5 ex> systemAliveCallbackSite =1|3	
alarmInCheck	Use alarm input check on/off	on | off ex> alarmInCheck =on	
alrmInCheckInterval	Alarm input check interval (hours)	1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 18 | 24 | 168 | 360 | 720 ex> alrmInCheckInterval =1	
alarmInCheckEmail	Use email action for alarm input check	on | off ex> alarmInCheckEmail=off	
alarmInCheckCallback	Use remote callback action for alarm input check	on | off ex> alarmInCheckCallback =on	
alarmInCheckCallbackSite	Use remote callback sites for alarm input check	1 | 2 | 3 | 4 | 5 ex> alarmInCheckCallbackSite =1|3	
sdCheck	Use SD card check on/off	on | off ex> sdCheck =on	
sdCheck kEmail	Use email action SD card check	on | off ex> sdCheckEmail=off	
sdCheckCallback	Use remote callback action for SD card check	on | off ex> sdCheckCallback =on	
sdCheckCallbackSite	Use remote callback sites for SD card check	1 | 2 | 3 | 4 | 5 ex> sdCheckCallbackSite =1|3	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=eventSystem&mode=0&systemAliveCheck=on&systemAliveInterval=1&systemAliveEmail=on&systemAliveCallback=on&systemAliveCallbackSite=1|2&alarmInCheck=on&alarmInCheckInterval=2&alarmInCheckEmail=off&alarmInCheckCallback=off&sdCheck=off

Response data:

returnCode=0

45. Lens Reset (for MFZ lens)

45.1. Write

Excute Lens reset command

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	lensReset	
mode	Mode Code	0	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action= lensReset&mode=0

Response data:

returnCode=0

46. Pan/Tilt Driver

46.1. Read

Get Pan/Tilt driver setup information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	pantiltDriver	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
driverName	Pan/Tilt Driver model name	Refer driverName ex> driverName=	
driverID	Pan/Tilt Driver ID	0~999 ex> driverID=0	
baudRate	Pan/Tilt Driver port speed	300 | 600 | 1200 | 2400 | 4800 | 9600 | 19200 | 38400 | 57600 | 115200 ex> baudRate=9600	
dataBit	Pan/Tilt Driver port data bit	7 | 8 ex> dataBit=8	
stopBit	Pan/Tilt Driver port stop bit	1 | 2 ex> stopBit=1	
parity	Pan/Tilt Driver port parity	none | odd | even ex>parity=none	

Example

Send data:
action=videoMisc&mode=1

Response data:
returnCode=0&videoOutType=ntsc

46.2. **Write**

Set Pan/Tilt driver setup information.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	pantiltDriver	
mode	Mode Code	0	
driverName	Pan/Tilt Driver model name	Refer driverName ex> driverName=	
driverID	Pan/Tilt Driver ID	0~999 ex> driverID=0	
baudRate	Pan/Tilt Driver port speed	300 | 600 | 1200 | 2400 | 4800 | 9600 | 19200 | 38400 | 57600 | 115200 ex> baudRate=9600	
dataBit	Pan/Tilt Driver port data bit	7 | 8 ex> dataBit=8	
stopBit	Pan/Tilt Driver port stop bit	1 | 2 ex> stopBit=1	
parity	Pan/Tilt Driver port parity	none | odd | even ex>parity=none	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=videoMisc&mode=0&videoOutType=ntsc

Response data:
returnCode=0

47. PTZ Capability

47.1. Read

Get PTZ capability information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzCapability	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
supportPTZ	Support PTZ functions	on | off ex> supportPTZ=on	
supportPan	Support pan functions	on | off ex> supportPan=on	
supportTilt	Support tilt functions	on | off ex> supportTilt=on	
supportZoom	Support zoom functions	on | off ex> supportZoom=on	
supportFocus	Support focus function.	on | off ex> supportFocus=on	
supportIris	Support iris functions	on | off ex> supportIris=off	
supportPreset	Support PTZ preset functions	on | off ex> supportPreset=on	
maxPresetCount	Max PTZ preset number	0 ~ 256 ex> maxPresetCount=0	

Example

Send data:
action=ptzCapability&mode=1

Response data:
returnCode=0&supportPTZ=on&supportPan=off&supportTilt=off&supportZoom=on&supportFocus=on&supportIris=off&supportPreset=on&maxPresetCount=16

48. PTZ Command

48.1. Control

Control PTZ functions.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzCommand	
mode	Mode Code	0	
command	PTZ commands	zoomIn | zoomOut | zoomStop | focusFar | focusNear | focusStop | focusOnepush | irisOpen | irisClose | irisStop | setIris moveToN | moveToNE | moveToE | moveToSE | moveToS | moveToSW | moveToW | moveToNW | stop | lensReset | wiperOn ex> command=focusOnepush	
step	Moving steps	<integer> 0 : continuous move, other move one step ex> step=0	
speed	Moving speed	1 ~ 16 ex> speed=4	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=ptzCommand&mode=0&command=zoomIn&step=0&speed=1

Response data:
returnCode=0

49. PTZ Command – TCP (for model group 1 & model group 2 only)

49.1. Read

read TCP port information for PTZ control

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzTcpControl	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
useServer	enable PTZ control server	on | off ex> useServer=on	
serverPort	PTZ control port	<interger> ex> serverPort=8015	
ipAddress{N}	allowed ip address for ptz control	1 ~ 10 ex> ipAddress=10.0.17.123	
{N} : 1 ~ 5			

Example

Send data:
action=ptzTcpControl&mode=1

Response data:
returnCode=0&userServer=on&serverPort=8015&ipAddress1=10.0.17.123

49.2. **Write**

write TCP port information for PTZ control

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzTcpControl	
mode	Mode Code	0	
useServer	enable PTZ control server	on | off ex> useServer=on	
serverPort	PTZ control port	<interger> ex> serverPort=8015	
registerMode	Set mode	add | remove ex> registerMode=add	
ipAddress	allowed ip address for ptz control	<IPAddress> ex> ipAddress=10.0.17.123	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=ptzTcpControl&mode=0&useServer=on&serverPort=8015&registerMode=add&ipAddress=10.0.17.123

Response data:
returnCode=0

50. PTZ Preset Command

50.1. Read

Control PTZ preset functions.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzPreset	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
presetName{N}	preset name	<string> ex> presetName1=door	
N : 1 ~ 256			

Example

Send data:
action=ptzPreset&mode=1

Response data:
returnCode=0&presetName1=door&presetName2=window

50.2. **Write**

Control PTZ preset functions.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzPreset	
mode	Mode Code	0	
command	PTZ preset commands	set | moveTo | remove ex> command=set	
id	Preset ID	1 ~ 256 ex> id=1	
presetName	preset name	<string> ex> presetName=door	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=ptzPreset&mode=0&command=set&id=1&presetName=door

Response data:

returnCode=0

Send data:

action=ptzPreset&mode=0&command=moveTo&id=1

Response data:

returnCode=0

51. PTZ Scan Command

51.1. Read

Control PTZ preset functions.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzScan	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
scanName{N}	PTZ scan Name	<string> ex> scanName1=path1	
startPreset{N}	PTZ scan start preset ID	1 ~ 255 ex> startPreset1=1	
endPreset{N}	PTZ scan end preset ID	1~255 ex> endPreset1=2	
dwellTime{N}	dwell time(sec) for PTZ scan operation	1~3600 ex>dwellTime=5	
speed{N}	speed for PTZ scan operation	1 ~ 16 ex> speed=1	
direction{N}	direction for PTZ scan operatio	clockwise | counterclockwise ex> direction=clockwise	
N : 1 ~ 8			

Example

Send data:
action=ptzPreset&mode=1

Response data:
returnCode=0&scanName1=path1&startPreset1=1&endPreset1=2&dwellTime1=1000&speed1=1&direction1=clockwise

51.2. **Write**

Control PTZ preset functions.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzScan	
mode	[Mode Code]()	0	
command	PTZ scan commands	set | remove | run | stop ex> command=setScan	
id	PTZ scan ID	1 ~ 8 ex> id=1	
scanName	PTZ scan Name (require only set command)	<string> ex> scanName=path1	
startPreset	PTZ scan start preset ID (require only set command)	1 ~ 255 ex> startPreset=1	
endPreset	PTZ scan end preset ID (require only set command)	1~255 ex> endPreset=2	
dwellTime	dwell time(sec) for PTZ scan operation (require only set command)	1~3600 ex>dwellTime=5	
speed	speed for PTZ scan operation (require only set command)	1 ~ 16 ex> speed=1	
direction	direction for PTZ scan operatio (require only set command)	clockwise | counterclockwise ex> direction=clockwise	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer [Return Code]()	

Example

Send data:
action=ptzPreset&mode=0&command=setPreset&id=1&presetName=door

Response data:
returnCode=0

Send data:
action=ptzPreset&mode=0&command=moveToPreset&id=1

Response data:
returnCode=0

52. PTZ Pattern Command

52.1. Read

read PTZ pattern information

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzPattern	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
patternName{N}	PTZ pattern Name	<string> ex> patternName1=path1	
N : 1 ~ 8			

Example

Send data:

action=ptzPattern&mode=1

Response data:

returnCode=0&patternName1=path1

52.2. **Control**

Control PTZ pattern functions.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzPattern	
mode	Mode Code	0	
command	PTZ pattern commands	recordStart | recordStop | run | stop | remove ex> command=recordStart	
id	PTZ pattern ID	1 ~ 8 ex> id=1	
patternName	PTZ pattern Name (require only recordStart command)	<string> ex> patternName=path1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=ptzPattern&mode=0&command= recordStart &id=1&patternName=pattern1

Response data:
returnCode=0

Send data:
action=ptzPattern&mode=0&command=recordStop&id=1

Response data:
returnCode=0

53. PTZ Tour Command

53.1. Read

Control PTZ preset functions.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzTour	
mode	Mode Code	1	
id	PTZ pattern ID	1 ~ 8 | all ex> id=1	

Receive Parameters (ID = all)

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
tourName{N}	PTZ tour Name	<string> ex> tourName1=path	
N : 1 ~ 8			

Example

Send data:
action=ptzTour&mode=1&id=all

Response data:
returnCode=0&tourName1=path

Receive Parameters (ID = 1~8)

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
use	use pattern	on | off	
function{N}	function for PTZ pattern	preset1~preset256 | scan1 ~ scan8 | pattern1~ pattern8 ex>function1=preset1 function2=scan1 function3=pattern1	
dwellTime{N}	dwell time(sec) for PTZ pattern operation	1~600 ex>dwellTime=5	
speed{N}	speed for PTZ pattern operation	1 ~ 16 ex> speed=1	
N : 1 ~ 8			

Example

Send data:
action=ptzTour&mode=1&id=1

Response data:
returnCode=0&use=on&function1=preset1&dwellTime1=5&speed1=1&function2=scan1&dwellTime2=5&speed2=8

53.2. **Write**

Control PTZ preset functions.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzTour	
mode	Mode Code	0	
command	PTZ tour commands	set | remove | run | stp[ex> command=set	
id	PTZ tour ID	1 ~ 8 ex> id=1	
tourName	name for PTZ tour (require only set command	<string> ex> tourName=tournSample	
function{N}	command set for tour (require only set command	preset1~preset256 | scan1 ~ scan8 | pattern1~ pattern8 ex>function1=preset1 function2=scan1 function3=pattern1	
dwellTime{N}	dwell time (sec) for next function (require only set command	1~600 ex>dwellTime=5	
speed{N}	speed for next function (require only set command	1 ~ 16 ex> speed=1	
N : 1 ~ functionCount			

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=ptzTour&mode=0&command=set&id=1&functionCount=0

Response data:
returnCode=0

Send data:
action=ptzTour&mode=0&command=remove&id=1

Response data:
returnCode=0

54. PTZ miscellaneous

54.1. Read

miscellaneout setup for PTZ

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzMisc	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
focusMode	set focus mode	auto | manual | onepush ex> focusMode=auto	
homePositionMode	set home position mode	default | userSetting | preset1~preset256 ex>homePositionMode=default	
autoRun	set auto run mode	none | preset1~preset256 | scan1~scan8 | pattern1~pattern8 | tour1~tour8 ex>autoRun=preset	
autoRunWaitTime	set auto run delay time(sec)	1~3600 ex> autoRunWaitTime=3	
autoPen	set auto pen	clockwise | counterclockwise ex>autoPen=clockwise	
autoFlip	set auto flip	off | digital | mechanical autoFlip=off	
tiltRange	set tilt range	90 | 100 ex> tiltRange=100	
restoreLastPosition	set restore last position	on | off ex> restoreLastPosition=off	

Example

Send data:
action= ptzMisc&mode=1

Response data:
returnCode=0

54.2. **Write**

miscellaneout setup for PTZ

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzMisc	
mode	Mode Code	0	
focusMode	set focus mode	auto | manual | onepush ex> focusMode=auto	
homePositionMode	set home position mode	default | userSetting | preset1~preset256 ex>homePositionMode=default	
autoRun	set auto run mode	none | preset1~preset256 | scan1~scan8 | pattern1~pattern8 | tour1~tour8 ex>autoRun=preset	
autoRunWaitTime	set auto run delay time(sec)	1~3600 ex> autoRunWaitTime=3	
autoPen	set auto pen	clockwise | counterclockwise ex>autoPen=clockwise	
autoFlip	set auto flip	off | digital | mechanical autoFlip=off	
tiltRange	set tilt range	90 | 100 ex> tiltRange=180	
restoreLastPosition	set restore last position	on | off ex> restoreLastPosition=off	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action= ptzMisc&mode=0&command=set&id=1&functionCount=0

Response data:
returnCode=0

55. PTZ Status

55.1. Read

Get ptz status.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzStatus	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
iris	iris position (optional)	0~100 ex> iris=100	
irisF	iris F number	<F number> ex> irisF=1.6	

Example

Send data:
action=ptzStatus&mode=1

Response data:
returnCode=0&iris=20&irisF=1.6

56. PTZ Absolute Move

56.1. Read

Get ptz absolute position

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzAbsolute	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
absPan	current pan position	-18000 ~ 18000 (unit 0.01°) ex> absPan=13500 (135°)	
absTilt	current tilt position	-9000 ~ 9000 (unit 0.01°) ex> absTilt=4500 (45°)	
absZoom	current zoom position	100 ~ <Max zoom sacle> (unit 0.01°) ex> absZoom=3000 (x30)	

Example

Send data:

action=ptzAbsolute&mode=1

Response data:

returnCode=0&absPan=18000&absTilt=8850&absZoom=3000

56.2. **Write**

Set ptz absolute position

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzAbsolute	
mode	Mode Code	0	
absPan	set pan position	-18000 ~ 18000 (unit 0.01º) ex> absPan=13500 (135º)	
absTilt	set tilt position	-9000 ~ 9000 (unit 0.01º) ex> absTilt=4500 (45º)	
absZoom	set zoom position	100 ~ <Max zoom sacle> (unit 0.01º) ex> absZoom=3000 (x30)	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=ptzAbsolute&mode=0&absPan=9000&absTilt=4500&absZoom=1000

Response data:

returnCode=0

57. PTZ Absolute Move Ex (Focus Position Extended)

57.1. Read

Get ptzf absolute position

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzAbsoluteEx	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
absPan	current pan position	-18000 ~ 18000 (unit 0.01º) ex> absPan=13500 (135º)	
absTilt	current tilt position	-9000 ~ 9000 (unit 0.01º) ex> absTilt=4500 (45º)	
absZoom	current zoom position	100 ~ <Max zoom sacle> (unit 0.01º) ex> absZoom=3000 (x30)	
absFocus	current focus position	0 ~ 10000 (range of focus lens to match the zoom scale)	

Example

Send data:

action=ptzAbsoluteEx&mode=1

Response data:

returnCode=0&absPan=18000&absTilt=8850&absZoom=3000&absFocus=5000

57.2. **Write**

Set ptzf absolute position

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzAbsoluteEx	
mode	Mode Code	0	
absPan	set pan position	-18000 ~ 18000 (unit 0.01°) ex> absPan=13500 (135°)	
absTilt	set tilt position	-9000 ~ 9000 (unit 0.01°) ex> absTilt=4500 (45°)	
absZoom	set zoom position	100 ~ <Max zoom sacle> (unit 0.01°) ex> absZoom=3000 (x30)	
absFocus	set focus position	0 ~ 10000 (range of focus lens to match the zoom scale) ex> absFocus=5000	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=ptzAbsoluteEx&mode=0&absPan=9000&absTilt=4500&absZoom=1000&absFocus=5000

Response data:

returnCode=0

58. PTZ Move to Point

58.1. Write

Move to point using ptz relative position.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzMoveToPoint	
mode	Mode Code	0	
pointPan	set pan point position	0 ~ 100000 (0.0 ~ 1.0: Relative coordinates, unit: 0.00001) ex> pointPan=50000 (center on image width)	
pointTilt	set tilt point position	0 ~ 100000 (0.0 ~ 1.0: Relative coordinates, unit: 0.00001) ex> pointPan=50000 (center on image height)	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=ptzMoveToPoint&mode=0&pointPan=20000&pointTilt=20000

Response data:
returnCode=0

59. PTZ Move to Area

59.1. Write

Move to area using ptz relative position.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	ptzMoveToArea	
mode	Mode Code	0	
left	set left point of area	0 ~ 100000 (0.0 ~ 1.0: Relative coordinates, unit: 0.00001) ex> left=50000 (center on image width)	
top	set top point of area	0 ~ 100000 (0.0 ~ 1.0: Relative coordinates, unit: 0.00001)	
right	set right point of area	0 ~ 100000 (0.0 ~ 1.0: Relative coordinates, unit: 0.00001)	
bottom	set bottom point of area	0 ~ 100000 (0.0 ~ 1.0: Relative coordinates, unit: 0.00001)	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=ptzMoveToArea&mode=0&left=200&top=200&right=90000&bottom=90000

Response data:

returnCode=0

60. Lens List

60.1. Read

Get lens list

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	lensList	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
lensCount	Number of lens	<integer> ex> lensCount=2	
lens{N}Name	lens name	<string> ex> lens1Name=idispl2	
lens{N}ID	lens setup ID	<interger> ex> lens1ID=0	
{N} : 1 ~ userCount			

Example

Send data:

action= lensList&mode=1

Response data:

returnCode=0& lensCount=2&lens1Name=idispl2&lens1ID=0&lens2Name=hdpropl1&lens1ID=1

61. Lens Driver

61.1. Read

Get lens driver setting information

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	lensDriver	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
lensName	lens name	<string> ex> lensName=idispl2	
minIris	minimum iris value	0~100 ex> minIris=0	
maxIris	maximum iris value	0~100 ex> maxIris=100	

Example

Send data:
action= lensDriver&mode=1

Response data:
returnCode=0&lensName=idispl2&minIris=0&maxIris=100

61.2. **Write**

Set lens driver setting information

Method: GET

Syntax
http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	lensDriver	
mode	Mode Code	0	
lensName	lens name	<string> ex> lensName=idispl2	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action= lensDriver&mode=1&lensName=idispl2

Response data:
returnCode=0

62. System Status

62.1. Read

Get system status.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	systemStatus	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
alarmIn	Alarm-in status	disable | on/off ex> alarmIn =on	
motionDetection	Motion status	disable | on/ off ex> motionDetection=on	
tripZone	Tripzone status	disable | on/ off ex> tripZone=on	
audioDetection	Audion detection status	disable | on/off ex> audioDetection=on	
tampering	Tempering status	disable | on/off ex> tampering=on	
faceDetection	Face detection status	disable | on/off ex> faceDetection=on	
alarmOut	Alarm-out status	disable | on/off ex> alarmOut=on	
alarmInHealth	Alarm-in health status	disable | on/off ex> alarmInHealth=on	
sdCard	SD status	disable/good/bad ex> sdCard=good	
recording	SD recording status	on | off ex> recording=off	
sdDataStartTime	SD recoding data start time	<date time > ex> sdDataStartTime=2013/12/03_09:00:01	
sdDataEndTime	SD recoding data end time	<date time > ex> sdDataEndTime=2013/12/03_15:12:34	

Example

Send data:
action=systemStatus&mode=1

Response data:
returnCode=0&alarmIn=off&motionDetection=on&tripZone=disable&audioDetection=disable&tampering=off&alarmOut=disable&alarmInHealth=on&sdCard=good&recording=off&sdDataStartTime=2013/12/03_09:00:01&sdDataEndTime=2013/12/03_15:12:34

63. Event Notification

63.1. Read

Get event notification server information.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventNotification	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
serverIP{N}	Server address	<IPv4 address> ex> serverIP1=10.0.1.2	
port{N}	Server port	<integer> ex> port1=1234	
timeOut{N}	network disconnect time when event doesn't exist (second)	<integer> 0~3600 ex> timeOut=0	Support Only group1, group2.
{N} : 1 ~ 5			

Example

Send data:
action=eventNotification&mode=1

Response data:
returnCode=0&serverIP1=10.0.1.2&port1=1234&timeOut1=0

TCP Event Message format

TCP packet is based string, It is sent as follows:

date=<value of YYYY/MM/DD format>&**time=**<value of HH:MM:SS > &**channel=**<event source channel>&**type=**<event type> *<CRLF>*

ex) date=2020/05/27&time=15:24:34&channel=0&type=alarm_in_on *<CRLF>*

- **date, time**: Time of event occurrence.
- **channel**: The channel where the event occurred. For example, if this camera is supports 2 alarm in device, total channel number is 2, channel increase zero based.
- **type** : type means event class and has the following values : (For details, refer to the Appendix 71. TCP Event Type) alarm_in_on, alarm_in_off, motion_on, motion_off, tripzone_on, tripzone_off ….

63.2. **Write**

Set event notification server information.
Up to 5 server settings are allowed. Settings are saved even after reboot (over WebAPI v2.15)
Parameters:timeOut is obsolete. (over WebAPI v2.15)

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventNotification	
mode	Mode Code	0	
registerMode	Server register mode	add | remove ex> registerMode=add	
serverIP	Server address	<IPv4 address> ex> serverIP=10.0.1.2	
port	Server port	<integer> ex> port=1234	
timeOut	network disconnect time when event doesn't exist (second)	<integer> 0~3600 ex> timeOut=0	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=eventNotification&mode=0&registerMode=add&serverIP=10.0.126.2&port=1234&timeOut=0

Response data:
returnCode=0

64. Event Log

64.1. Read

Get event logs.

Method: GET

Syntax
http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	eventLog	
mode	Mode Code	1	
logCount	Number of event logs	1 ~ 32 ex> logCount=5	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
log{N}	The latest {N} event logs	YYYY/MM/DD_hh:mm:ss_<event name>_off | YYYY/MM/DD_hh:mm:ss_<event name>_on <event name> = alarmIn | motion | tripZone | audio | tampering ex> log1=2014/02/14_12:34:56_motion_off log2=2014/02/14_12:33:56_audio_on log3=2014/02/14_11:33:56_tripZone_off	
{N} : 1 ~ logCount			

Example

Send data:
action=eventLog&mode=1&logCount=3

Response data:
returnCode=0&log1=2014/02/14_12:34:56_motion_off&log2=2014/02/14_12:33:56_motion_on&log3=2014/02/14_11:33:56_motion_off

65. System Log

65.1. System log Export

convert system log to txt, it return txt file url

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	systemLog	
mode	Mode Code	1	
beginPos	Begin Position	1 ~ 5000, beginPos <= endPos <= logCount	
endPos	End Position	1 ~ 5000, beginPos <= endPos <= logCount	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
logFileURL	URL for converted txt file	<string> ex> logFileURL=http://10.0.10.70/systemlog.txt	

Example

Send data:

action=systemLog&mode=1

Response data:

returnCode=0&logFileURL=http://10.0.126.70/systemlog.txt

65.2. **Read**

Get System Logs

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	systemLog	
mode	Mode Code	4	
beginPos	Begin Position	1 ~ 5000, beginPos <= endPos <= logCount	
endPos	End Position	1 ~ 5000, beginPos <= endPos <= logCount	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
logCount	Number of logs	1 ~ 5000	
log	log	<string>	

Example

Send data:

action=systemLog&mode=4&beginPos=1&endPos=300

Response data:

returnCode=0&logCount=5000&log= [2022/11/04_17:28:07]
type=10000000, data=
type=16000000, data=
type=16001000, data=[MEDIA] mediaStream 0 connected with streamKey oefm8odsoqqbagd5hh0hi1p50ro9cvvq

[2022/11/04_17:28:30] Remote User Logout ([RTSP @110.234.154.139 1])
…

66. Debug Log

66.1. Debug log Export

convert debug log to txt, it return txt file url

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	debugLog	
mode	Mode Code	1	
beginPos	Begin Position	1 ~ 5000, beginPos <= endPos <= logCount	
endPos	End Position	1 ~ 5000, beginPos <= endPos <= logCount	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
logFileURL	URL for converted txt file	<string> ex> logFileURL=http://10.0.10.70/debuglog.txt	

Example

Send data:
action=debugLog&mode=1

Response data:
returnCode=0&logFileURL=http://10.0.126.70/debugLog.txt

66.2. **Read**

Get Debug Logs

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	debugLog	
mode	Mode Code	4	
beginPos	Begin Position	1 ~ 5000, beginPos <= endPos <= logCount	
endPos	End Position	1 ~ 5000, beginPos <= endPos <= logCount	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
logCount	Number of logs	1 ~ 5000	
log	log	<string>	

Example

Send data:

action=debugLog&mode=4&beginPos=1&endPos=300

Response data:

returnCode=0&logCount=5000&log=[2022/11/03_11:38:39] [RTSP][C][11920]count=2 : ,
[2022/11/03_11:39:02] [MEDIA] onRtpStreamingCompleted stream 0
[2022/11/03_11:39:02] [RTSP][D][11920]count=1 : ,   reason:1
[2022/11/03_11:40:03] [RTSP][C][11941]count=2 : ,
[2022/11/03_11:40:03] [RTSP][D][11941]count=1 : ,   reason:1
[2022/11/03_11:40:03] [RTSP][C][11942]count=2 : ,
[2022/11/03_11:40:24] [MEDIA] onRtpStreamingCompleted stream 0
[2022/11/03_11:40:24] [RTSP][D][11942]count=1 : ,   reason:1
[2022/11/03_11:40:49] [RTSP][C][11956]count=2 : ,
[2022/11/03_11:40:49] [RTSP][D][11956]count=1 : ,   reason:1
…

67. Fisheye (for model group 4 only)

67.1. Read

Read Fisheye setting functions.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	fishEye	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
intelligentCodec	Use intelligent codec	on | off ex> intelligentCodec=on	
useDewarp	Use Dewarping mode	on | off ex> useDewarp=on	
mountType	Mount type	wall | desktop | ceiling | ex> mountType=wall	
mainScreenSize	Main screen size	2048x2048 | 1024x1024 | 2048x1024 | 1024x512 ex> mainScreenSize=2048x2048	
screenType	Fisheye screen type	1R | 1P | 1P2R | 4R | 2P | 8R ex> screenType=4R	
subRegionName{N}	Sub region name	<string> ex> subRegionName1=door	
dewarpType{N}	Dewarp type	normal | subRegion | panorama ex> dewarpType1=subRegion	
tiltValue{N}	Tilt value	-90 ~ 90 ex> tiltValue1=45	
panValue{N}	Pan value	-180~180 ex> panValue1=90	
N : 1 ~ 8			
useStream{M}	Use N-th stream	on | off ex> useStream1=on	
codecStream{M}	Stream encoding CODEC	h264 | jpeg ex> codecStream1=h264	
subRegionStream{M}	Stream target subregion when set dewarp on	1 ~ 8 | all ex> subRegionStream1=1	
resolutionStream{M}	Stream resolution when set dewarp off	2048x2048 | 1024x1024 | 512x512 ex> resolutionStream =2048x2048	
qualityStream{M}	Stream quality	basic | standard | high | veryHigh ex> qualityStream1=veryHigh	
bitrateControlStream{M}	Bitrate control mode	vbr | cbr ex> bitrateControlStream1=cbr	

framerateStream{M}	Stream frame per second	30 | 15 | 10 | 5 | 4 | 3 | 2 | 1 ex> framerateStream1=30	
M : 1 ~ 6			

Example

Send data:

action=fisheye&mode=1

Response data:

returnCode=0&intelligentCodec=on&useDewarp=on&dewarpType1=normal&dewarpType2=subRegion&tiltValue2=65&panValue2=180&dewarpType3=subRegion&tiltValue3=65&panValue3=-129&dewarpType4=subRegion&tiltValue4=65&panValue4=-77
&dewarpType5=subRegion&tiltValue5=65&panValue5=-26&dewarpType6=subRegion&tiltValue6=65&panValue6=26
&dewarpType7=subRegion&tiltValue7=65&panValue7=77&dewarpType8=subRegion&tiltValue8=65&panValue8=129
&mountType=desktop&mainScreenSize=&screenType=8R
&subRegionStream1=1&qualityStream1=standard&useStream1=on&codecStream1=h264&bitrateControlStream1=vbr&framerateStream1=30&resolutionStream1=
&subRegionStream2=all&qualityStream2=standard&useStream2=off&codecStream2=h264&bitrateControlStream2=vbr&framerateStream2=30&resolutionStream2=
&subRegionStream3=all&qualityStream3=standard&useStream3=off&codecStream3=h264&bitrateControlStream3=vbr&framerateStream3=0&resolutionStream3=
&subRegionStream4=all&qualityStream4=standard&useStream4=off&codecStream4=h264&bitrateControlStream4=vbr&framerateStream4=0&resolutionStream4=
&subRegionStream5=all&qualityStream5=standard&useStream5=off&codecStream5=h264&bitrateControlStream5=vbr&framerateStream5=0&resolutionStream5=
&subRegionStream6=all&qualityStream6=standard&useStream6=off&codecStream6=h264&bitrateControlStream6=vbr&framerateStream6=0&resolutionStream6=

67.2. **Write**

Write Fisheye setting functions.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	**Description**	**Values**	**ETC**
action	Action ID	fishEye	
mode	Mode Code	0	
intelligentCodec	Use intelligent codec	on | off ex> intelligentCodec=on	
useDewarp	Use Dewarping mode	on | off ex> useDewarp=on	
mountType	Mount type	wall | desktop | ceiling | ex> mountType=wall	
mainScreenSize	Main screen size	2048x2048 | 1024x1024 | 2048x1024 | 1024x512 ex> mainScreenSize=2048x2048	
screenType	Fisheye screen type	1R | 1P | 1P2R | 4R | 2P | 8R ex> screenType=4R	
useStream{M}	Use N-th stream	on | off ex> useStream1=on	
codecStream{M}	Stream encoding CODEC	h264 | jpeg ex> codecStream1=h264	
subRegionStream{M}	Stream target subRegion when set dewarp on	1 ~ 8 | all ex> subRegionStream1=1	
resolutionStream{M}	Stream resolution when set dewarp off	2048x2048 | 1024x1024 | 512x512 ex> resolutionStream =2048x2048	
qualityStream{M}	Stream quality	basic | standard | high | veryHigh ex> qualityStream1=veryHigh	
bitrateControlStream{M}	Bitrate control mode	vbr | cbr ex> bitrateControlStream1=cbr	
framerateStream{M}	Stream frame per second	30 | 15 | 10 | 5 | 4 | 3 | 2 | 1 ex> framerateStream1=30	
M : 1 ~ 6			

Receive Parameters

Parameters	**Description**	**Values**	**ETC**
returnCode	Return code	Refer Return Code	

Example

Send data:

action=fisheye&mode=1&
intelligentCodec=on&useDewarp=on&dewarpType1=normal&dewarpType2=subRegion&tiltValue2=65&panValue2=180
&dewarpType3=subRegion&tiltValue3=65&panValue3=-129&dewarpType4=subRegion&tiltValue4=65&panValue4=-77
&dewarpType5=subRegion&tiltValue5=65&panValue5=-26&dewarpType6=subRegion&tiltValue6=65&panValue6=26
&dewarpType7=subRegion&tiltValue7=65&panValue7=77&dewarpType8=subRegion&tiltValue8=65&panValue8=129

```
&mountType=desktop&mainScreenSize=&screenType=8R
&subRegionStream1=1&qualityStream1=standard&useStream1=on&codecStream1=h264&bitrateControlStream1=vbr&framerateStream1=30&resolutionStream1=
&subRegionStream2=all&qualityStream2=standard&useStream2=off&codecStream2=h264&bitrateControlStream2=vbr&framerateStream2=30&resolutionStream2=
&subRegionStream3=all&qualityStream3=standard&useStream3=off&codecStream3=h264&bitrateControlStream3=vbr&framerateStream3=0&resolutionStream3=
&subRegionStream4=all&qualityStream4=standard&useStream4=off&codecStream4=h264&bitrateControlStream4=vbr&framerateStream4=0&resolutionStream4=
&subRegionStream5=all&qualityStream5=standard&useStream5=off&codecStream5=h264&bitrateControlStream5=vbr&framerateStream5=0&resolutionStream5=
&subRegionStream6=all&qualityStream6=standard&useStream6=off&codecStream6=h264&bitrateControlStream6=vbr&framerateStream6=0&resolutionStream6=
```

Response data:

returnCode=0

68. Fisheye (for model group 6 only)

68.1. Read

Read Fisheye setting functions.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	fishEye	
mode	Mode Code	1	

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	
intelligentCodec	Use intelligent codec	on | off ex> intelligentCodec=on	
useDewarp	Use Dewarping mode	on | off ex> useDewarp=on	
mountType	Mount type	wall | desktop | ceiling | ex> mountType=wall	
mainScreenSize	Main screen size	3200x3008 | 1600x1504 | 3200x1504 | 1600x752 ex> mainScreenSize=3200x3008	
screenType	Fisheye screen type	1R | 1P | 1P2R | 4R | 2P | 8R ex> screenType=4R	
subRegionName{N}	Sub region name	<string> ex> subRegionName1=door	
dewarpType{N}	Dewarp type	normal | subRegion | panorama ex> dewarpType1=subRegion	
tiltValue{N}	Tilt value	-90 ~ 90 ex> tiltValue1=45	
panValue{N}	Pan value	-180~180 ex> panValue1=90	
N : 1 ~ 8			
useStream{M}	Use N-th stream	on | off ex> useStream1=on	
codecStream{M}	Stream encoding CODEC	h265 | h264 | jpeg ex> codecStream1=h264	
subRegionStream{M}	Stream target subregion when set dewarp on	1 ~ 8 | all ex> subRegionStream1=1	
resolutionStream{M}	Stream resolution when set dewarp off	3200x3008 | 1600x1504 | 800x752 ex> resolutionStream1 = 800x752	
qualityStream{M}	Stream quality	basic | standard | high | veryHigh ex> qualityStream1=veryHigh	
bitrateControlStream{M}	Bitrate control mode	vbr | cbr ex> bitrateControlStream1=cbr	

framerateStream{M}	Stream frame per second	30 | 15 | 10 | 5 | 4 | 3 | 2 | 1 ex> framerateStream1=30	
M : 1 ~ 6			

Example

Send data:

action=fisheye&mode=1

Response data:

returnCode=0&intelligentCodec=on&useDewarp=on&dewarpType1=normal&dewarpType2=subRegion&tiltValue2=65&panValue2=180&dewarpType3=subRegion&tiltValue3=65&panValue3=-129&dewarpType4=subRegion&tiltValue4=65&panValue4=-77&dewarpType5=subRegion&tiltValue5=65&panValue5=-26&dewarpType6=subRegion&tiltValue6=65&panValue6=26&dewarpType7=subRegion&tiltValue7=65&panValue7=77&dewarpType8=subRegion&tiltValue8=65&panValue8=129&mountType=desktop&mainScreenSize=&screenType=8R&subRegionStream1=1&qualityStream1=standard&useStream1=on&codecStream1=h264&bitrateControlStream1=vbr&framerateStream1=30&resolutionStream1=&subRegionStream2=all&qualityStream2=standard&useStream2=off&codecStream2=h264&bitrateControlStream2=vbr&framerateStream2=30&resolutionStream2=&subRegionStream3=all&qualityStream3=standard&useStream3=off&codecStream3=h264&bitrateControlStream3=vbr&framerateStream3=0&resolutionStream3=&subRegionStream4=all&qualityStream4=standard&useStream4=off&codecStream4=h264&bitrateControlStream4=vbr&framerateStream4=0&resolutionStream4=&subRegionStream5=all&qualityStream5=standard&useStream5=off&codecStream5=h264&bitrateControlStream5=vbr&framerateStream5=0&resolutionStream5=&subRegionStream6=all&qualityStream6=standard&useStream6=off&codecStream6=h264&bitrateControlStream6=vbr&framerateStream6=0&resolutionStream6=

68.2. **Write**

Write Fisheye setting functions.

Method: GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	fishEye	
mode	Mode Code	0	
intelligentCodec	Use intelligent codec	on | off ex> intelligentCodec=on	
useDewarp	Use Dewarping mode	on | off ex> useDewarp=on	
mountType	Mount type	wall | desktop | ceiling | ex> mountType=wall	
mainScreenSize	Main screen size	3200x3008 | 1600x1504 | 3200x1504 | 1600x752 ex> mainScreenSize=3200x3008	
screenType	Fisheye screen type	1R | 1P | 1P2R | 4R | 2P | 8R ex> screenType=4R	
useStream{M}	Use N-th stream	on | off ex> useStream1=on	
codecStream{M}	Stream encoding CODEC	h265 | h264 | jpeg ex> codecStream1=h264	
subRegionStream{M}	Stream target subRegion when set dewarp on	1 ~ 8 | all ex> subRegionStream1=1	
resolutionStream{M}	Stream resolution when set dewarp off	3200x3008 | 1600x1504 | 800x752 ex> resolutionStream1 = 800x752	
qualityStream{M}	Stream quality	basic | standard | high | veryHigh ex> qualityStream1=veryHigh	
bitrateControlStream{M}	Bitrate control mode	vbr | cbr ex> bitrateControlStream1=cbr	
framerateStream{M}	Stream frame per second	30 | 15 | 10 | 5 | 4 | 3 | 2 | 1 ex> framerateStream1=30	
M : 1 ~ 6			

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=fisheye&mode=1&
intelligentCodec=on&useDewarp=on&dewarpType1=normal&dewarpType2=subRegion&tiltValue2=65&panValue2=180
&dewarpType3=subRegion&tiltValue3=65&panValue3=-129&dewarpType4=subRegion&tiltValue4=65&panValue4=-77
&dewarpType5=subRegion&tiltValue5=65&panValue5=-26&dewarpType6=subRegion&tiltValue6=65&panValue6=26
&dewarpType7=subRegion&tiltValue7=65&panValue7=77&dewarpType8=subRegion&tiltValue8=65&panValue8=129

&mountType=desktop&mainScreenSize=&screenType=8R
&subRegionStream1=1&qualityStream1=standard&useStream1=on&codecStream1=h264&bitrateControlStream1=vbr&framerateStream1=30&resolutionStream1=
&subRegionStream2=all&qualityStream2=standard&useStream2=off&codecStream2=h264&bitrateControlStream2=vbr&framerateStream2=30&resolutionStream2=
&subRegionStream3=all&qualityStream3=standard&useStream3=off&codecStream3=h264&bitrateControlStream3=vbr&framerateStream3=0&resolutionStream3=
&subRegionStream4=all&qualityStream4=standard&useStream4=off&codecStream4=h264&bitrateControlStream4=vbr&framerateStream4=0&resolutionStream4=
&subRegionStream5=all&qualityStream5=standard&useStream5=off&codecStream5=h264&bitrateControlStream5=vbr&framerateStream5=0&resolutionStream5=
&subRegionStream6=all&qualityStream6=standard&useStream6=off&codecStream6=h264&bitrateControlStream6=vbr&framerateStream6=0&resolutionStream6=

Response data:
returnCode=0

69. Fisheye Command (for model group 4 & 6 only)

69.1. Read

Read Fisheye subregion setup

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	fishEyeCommand	
mode	Mode Code	1	
subRegionName{N}	Sub region name	<string> ex> subRegionName1=door	
dewarpType{N}	Dewarp type	normal | subRegion | panorama ex> dewarpType1=subRegion	
tiltValue{N}	Tilt value	-90 ~ 90 ex> tiltValue1=45	
panValue{N}	Pan value	-180~180 ex> panValue1=90	
N : 1 ~ 8			

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:
action=fishEyeCommand&mode=1

Response data:
returnCode=0&dewarpType1=normal&dewarpType2=subRegion&tiltValue2=65&panValue2=180
&dewarpType3=subRegion&tiltValue3=65&panValue3=-129&dewarpType4=subRegion&tiltValue4=65&panValue4=-77
&dewarpType5=subRegion&tiltValue5=65&panValue5=-26&dewarpType6=subRegion&tiltValue6=65&panValue6=26
&dewarpType7=subRegion&tiltValue7=65&panValue7=77&dewarpType8=subRegion&tiltValue8=65&panValue8=129

69.2. **Write**

Write Fisheye subregion setup.

Method: POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi

Send Parameters

Parameters	Description	Values	ETC
action	Action ID	fishEyeCommand	
mode	Mode Code	0	
subRegionName{N}	Sub region name	<string> ex> subRegionName1=door	
dewarpType{N}	Dewarp type	normal | subRegion | panorama ex> dewarpType1=subRegion	
tiltValue{N}	Tilt value	-90 ~ 90 ex> tiltValue1=45	
panValue{N}	Pan value	-180~180 ex> panValue1=90	
N : 1 ~ 8			

Receive Parameters

Parameters	Description	Values	ETC
returnCode	Return code	Refer Return Code	

Example

Send data:

action=fishEyeCommand&mode=0&dewarpType1=normal&dewarpType2=subRegion&tiltValue2=65&panValue2=180
&dewarpType3=subRegion&tiltValue3=65&panValue3=-129&dewarpType4=subRegion&tiltValue4=65&panValue4=-77
&dewarpType5=subRegion&tiltValue5=65&panValue5=-26&dewarpType6=subRegion&tiltValue6=65&panValue6=26
&dewarpType7=subRegion&tiltValue7=65&panValue7=77&dewarpType8=subRegion&tiltValue8=65&panValue8=129

Response data:

returnCode=0

70. Preparing File Upload

This action must be preceded before requesting out of all actions for uploading file.
(ex, firmware upgrade, audio file upload, import setup, certification file upload, etc)

Method : GET

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?action=uploadPrepare

Receive Parameters

Parameters	Description	Values	Target
returnCode	Returns approprieate value	0: File upload available 9999: Camera is already downloading something * returnCode must be ‘0’ before trying file upload	All

71. File Upload

Method : POST

Syntax

http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?<parameter>=<value>[&<parameter>=<value>…]

Receive Parameters

Parameters	Description	Values	Target
returnCode	Returns result of uploading	0: Upgrade succeed 101: Upgrade failed	All

* Example for using HTML form tag

```
==========================================================================================
<form enctype="multipart/form-data" action="http[s]://<device ip>:<port>/cgi-bin/webSetup.cgi?action=firmwareUpgrade"
method="post">
    <p> Firmware file</p>
    <input name="uploadedFile " type="file” >
    <br /><br />
    <input type="submit" value="start">
</form>
==========================================================================================
```

* Example for using packet

```
==========================================================================================
POST /cgi-bin/webSetup.cgi?action=firmwareUpgrade HTTP/1.1 <CRLF>
Content-Type: multipart/form-data; boundary=<boundaryString><CRLF>
Host: <device ip><CRLF>
Content-Length: 26477419<CRLF>
Connection: Keep-Alive<CRLF>
Cache-Control: no-cache<CRLF>
Authorization: Digest username="admin",realm="IDIS Web Server",nonce="400010b0ca535dd820d0",uri="/cgi-
bin/webSetup.cgi?action=firmwareUpgrade",cnonce="3c73bac9a962c8a73e59c54bad07ef0f",nc=00000002,response="2f55557e5
af7029c7470ca7ece3a3c6a",qop="auth",opaque="0c8ee46f"<CRLF>
<CRLF>
<boundaryString><CRLF>
Content-Disposition: form-data; name="uploadedFile"; filename="X:\S1_PROCOTOL\release\252\ncea220wp2-s1-kor-100-
org2006_dip.rui"<CRLF>
Content-Type: application/octet-stream<CRLF>
<CRLF>
…………………………………………………………………………………….
…………………………..[binary data]……………………………………….
…………………………………………………………<CRLF>
<boundaryString><CRLF>
==========================================================================================
```

72. Motion Block Information

Description of block data

	Block width	Separator	Block Height	Separator	Block data
	4 Digit decimal number	"_"	4 Digit decimal number	"_"	A block data is denoted by 1 bit. Each row is the hexadecimal notation. 0:off 1:on
ex)	0026	_	0015	_	

When describing block data, put "_" at the end of each row.
If the block width is not a multiple of 4, fill up a dummy value to meet multiple of 4 align.

ex) 9x5 block

3 bit	2 bit	1 bit	0bit	3 bit	2 bit	1 bit	0bit	3 bit	2 bit	1 bit	0bit	hexadecimal
0	0	0	0	0	0	0	0	0	0(dummy)	0(dummy)	0(dummy)	0x000
0	0	1	1	1	0	0	0	0	0(dummy)	0(dummy)	0(dummy)	0x380
0	0	1	1	1	1	0	0	0	0(dummy)	0(dummy)	0(dummy)	0x3C0
0	0	0	1	1	1	1	0	0	0(dummy)	0(dummy)	0(dummy)	0x1E0
0	0	0	0	1	1	1	1	1	0(dummy)	0(dummy)	0(dummy)	0x0f8

⇨ 0009_0005_000_380_3c0_1e0_0f8

ex) 16x5 block

3 bit	2 bit	1 bit	0bit	3 bit	2 bit	1 bit	0bit	3 bit	2 bit	1 bit	0bit	3 bit	2 bit	1 bit	0bit	hexadecimal
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x0000
0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0x0380
0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0x03C0
0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0x01E0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0	0x00f8

⇨ 0016_0005_0000_0380_03c0_01e0_00f8

73. Schedule Information

Description of schedule information

8Bits indicated as 1 hour. i.e. two bits to indicate 15 minutes. Monday to Sunday from 0:00 to 23:59, fill up the schedule information for each column.

put "_" at each day of a week.

ex) The schedule that you want to convert

		HOUR																							
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
DAY	Mo																								
	Tu									■	■	■	■	■	■	■	■								
	We									■	■	■	■	■	■	■	■								
	Th									■	■	■	■	■	■	■	■								
	Fr									■	■	■	■	■	■	■	■								
	Sa																								
	Su																								

Zoom in Tuesday schedule information

Hour	0				1				2				3				4				5			
Min	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45
2bits	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
Hex	0		0		0		0		0		0		0		0		0		0		0		0	
Hour	6				7				8				9				10				11			
Min	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45
											■	■	■	■	■	■	■	■	■	■	■	■	■	■
2bits	00	00	00	00	00	00	00	00	00	00	01	01	01	01	01	01	01	01	01	01	01	01	01	01
Hex	0		0		0		0		0		5		5		5		5		5		5		5	
Hour	12				13				14				15				16				17			
Min	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45
	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■									
2bits	01	01	01	01	01	01	01	01	01	01	01	01	01	01	01	00	00	00	00	00	00	00	00	00
Hex	5		5		5		5		5		5		5		3		0		0		0		0	
Hour	18				19				20				21				22				23			
Min	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45	0	15	30	45
2bits	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
Hex	0		0		0		0		0		0		0		0		0		0		0		0	

⇨ 000000000000000005555555555555530000000000000000

The converted data stream for the above schedule information

⇨ 000000000000000000000000000000000000000000000000_000000000000000005555555555555530000000000000000_000000000000000555555555555555550000000000000000_000000000000000555555555555555550000000000000000_000000000000000555555555555555550000000000000000_000000000000000000000000000000000000000000000000_000000000000000000000000000000000000000000000000

74. Mode Code

Mode Code	Description
0	Write
1	Read
2	Test
5	Secured Write
6	Secured Read
7	Secured Test
700	F/W upgrade
702	System Factory Default
9000	System Restart
9001	Upload file
9002	Upload file in web UI (*)

75. Return Code

Type	Return Code	Description
Success	0	Success
Error	100	File does not exist.
	101	Firmware upgrade fail
	102	Failed to create the system data
	107	Failed to send email
	108	System wizard failed
	109	Network wizard failed
	110	DDNS failed
	111	camera name already exist in DDNS server
	200	Failed to mount SD card
	201	Failed to unmount SD card
	202	Failed to format SD card
	203	Failed to check SD card
	204	Failed to mount SD card (already mounted)
	205	Failed to format SD card (unmounted SD Card)
	301	Invalid parameter(s) (out of the range of valid values)
	302	Either attempting to write a read-only data or written data is different from the original data.
	303	Over the camera compression capability. Reduce the resolution or frame rates.
	304	Not enough parameters (need the mandatory parameters)
	305	INVALID_SETUP_FILE
	306	NOT_HAS_PERMISSION
	307	NOT_FOUND_SESSION_INFO
	308	NOT_SUPPORT_CMD_THIS_VERSION
	309	DATA_DECRYPT_FAILED
	310	FUNCTION_NOT_SUPPORTED
	400	UPNP NAT not exist
	401	UPNP error
	402	UPNP invalid port range
	500	FTP upload test failed
	600	SMTP_TEST_FAIL_UNKNOWN
	601	SMTP_TEST_FAIL_INVALID_DNS
	602	SMTP_TEST_FAIL_TO_CONNECT_SMTP
	603	SMTP_TEST_FAIL_TO_CONNECT_SSL
	604	SMTP_TEST_FAIL_NO_SUCH_USER
	605	SMTP_TEST_FAIL_TO_AUTHENTICATE
	606	SMTP_TEST_FAIL_NEED_TLS

	607	SMTP_TEST_FAIL_TO_SEND_MESSAGE
	608	SMTP_TEST_FAIL_TO_SEND_CMD_FROM
	609	SMTP_TEST_FAIL_TO_SEND_CMD_RCPT
	610	SMTP_TEST_FAIL_TO_CMD_DATA
	611	SMTP_TEST_FAIL_TO_CMD_DATA_END
	612	SMTP_TEST_FAIL_TO_CMD_QUIT
	900	Password authentication failed (wrong password)
	901	Password policy violation
	902	Exceed the maximum number of element (user, group, alarm file, PTZ client, preset, scan, tour, event client)
	903	No matching user found
	904	There is already same element(user, group, alarm file, PTZ client, preset, scan, tour, event client)
	9000	Unknown API
	9999	Unknown error

76. Time Zone

Description	GMT
HowlandIsland	GMT-12:00
MidwayIsland_Samoa	GMT-11:00
Hawaii	GMT-10:00
Alaska	GMT-09:00
PacificTime PacificTime_Tijuana	GMT-08:00
MountainTime Arizona Chihuahua_Mazatlan	GMT-07:00
MexicoCity Saskatchewan_Tegucigalpa CentralTime CentralAmerica	GMT-06:00
EasternTime Bogota_Lima_Quito Indianapolis	GMT-05:00
AtlanticTime Santiago Caracas_LaPaz_Georgetown	GMT-04:00
Newfoundland	GMT-03:30
Greenland BuenosAires Brasilia	GMT-03:00
MidAtlantic	GMT-02:00
CapeVerdeIs Azores	GMT-01:00
Dublin_Edinburgh_Lisbon_London Greenwich_Monrovia	GMT 00:00
Belgrade_Bratislava_Budapest_Ljubljana_Prague Brussels_Copenhagen_Madrid_Paris Sarajevo_Skopje_Warsaw_Zagreb WestCentralAfrica Amsterdam_Berlin_Bern_Rome_Stockholm_Vienna	GMT+01:00
Bucharest_Vilnius Athens Jerusalem Cairo Harare_Pretoria Helsinki_Riga_Sofia_Tallinn	GMT+02:00
Nairobi_Istanbul_Minsk Moscow_StPetersburg Baghdad_Kuwait_Riyadh Volgograd	GMT+03:00
Tehran	GMT+03:30
Baku_Tbilisi AbuDhabi_Muscat	GMT+04:00
Kabul	GMT+04:30
Ekaterinburg Islamabad_Karachi_Tashkent	GMT+05:00
Bombay_Calcutta_Madras_NewDelhi_SriJayawardenepura	GMT+05:30
Kathmandu	GMT+05:45
Thimphu_Urumqi Astana_Dhaka Almaty	GMT+06:00
Rangoon	GMT+06:30
Bangkok_Hanoi_Jakarta Novosibirsk_Krasnoyarsk	GMT+07:00
Beijing_Chongqing_HongKong Irkutsk_UlaanBataar Singapore Taipei Perth	GMT+08:00

Seoul Yakutsk Osaka_Sapporo_Tokyo	GMT+09:00
Darwin Adelaide	GMT+09:30
Guam_PortMoresby Brisbane Vladivostok Canberra_Melbourne_Sydney Hobart	GMT+10:00
Magadan_SolomonIs_NewCaledonia	GMT+11:00
Auckland_Wellington Kamchatka_Kwajalein_MarshallIs	GMT+12:00
Nukualofa	GMT+13:00

77. Pan/Tilt Driver model name

Description	Manufacturer	Model
VisionDome (360Vision)	360Vision	VisionDome
OrbiterMicrosphere (Ademcovideo)	Ademco video	OrbiterMicrosphere
VC-C4 (Canon)	Canon	VC-C4
VC-C50i (Canon)	Canon	VC-C50i
ZC-SD622J (CBC)	CBC	ZC-SD622J
CRD-J6416 (Chilsung)	Chilsung	CRD-J6416
CDC-2500 (Costar)	Costar	CDC-2500
SIC722V (Costar)	Costar	SIC722V
Dennard2060 (Dennard)	Dennard	Dennard2060
DY-255RXC (Dongyang)	Dongyang	DY-255RXC
PowerController (Dongyang)	Dongyang	PowerController
DRX-500 (Dongyang Unitech)	Dongyang Unitech	DRX-500
CDC-2400 (DynaColor)	DynaColor	CDC-2400
PTC-200C/CVAS (ELMO)	ELMO	PTC-200C/CVAS
PTC-250C (ELMO)	ELMO	PTC-250C
PTC-400 (ELMO)	ELMO	PTC-400
CRX-1401 (ERAESEEDS)	ERAESEEDS	CRX-1401
CRR-1660s (Fine)	Fine	CRR-1660s
Fastrax (HiTron)	HiTron	Fastrax
Fastrax2 (HiTron)	HiTron	Fastrax2
SpeedDome (HiTron)	HiTron	SpeedDome
22xAFZoom (Hitron)	HiTron	22xAFZoom
GRU604A (LG Honeywell)	Honeywell	GRU604A
HDC-655 (Honeywell)	Honeywell	HDC-655
HSD-25X (LG Honeywell)	Honeywell	HSD-25X
HSDN-251 (Honeywell)	Honeywell	HSDN-251
IRX-100 (IDIS)	IDIS	IRX-100
VRX-2201 (Inter-M)	Inter-M	VRX-2201
TK-S576 (JVC)	JVC	TK-S576
KTD-312 (Kalatel)	Kalatel	KTD-312
LGSpeedDome	LG	LGSpeedDome
LPT-A100L (LG)	LG	LPT-A100L
PIH-717 (Linlin)	Linlin	PIH-717
NIKO (NewBornHightech)	New Born Hightech	NIKO
NOVUS-C	NOVUS	NOVUS-C
N-Control (NOVUS)	NOVUS	N-Control

Pacom2036 (Pacom)	Pacom	Pacom2036
WJ-SX550A (Panasonic)	Panasonic	WJ-SX550A
WV-CS850/854 (Panasonic)	Panasonic	WV-CS850/854
D-protocol (Pelco)	Pelco	D-protocol
SpectraDome (Pelco)	Pelco	SpectraDome
G3BasicAutoDome (Philips)	Philips	G3BasicAutoDome
MRX-1000 (Samsung)	Samsung	MRX-1000
SamsungDome (Samsung)	Samsung	SamsungDome
ZoomCamera (Samsung)	Samsung	ZoomCamera
SPD1600 (SamsungTechwin)	Samsung Techwin	SPD1600
SRX-100B (SamsungTechwin)	Samsung Techwin	SRX-100B
DeltaDomeII/UltraIV (Sensormatic)	Sensormatic	DeltaDomeII/UltraIV
UltraVI (Sensormatic)	Sensormatic	UltraVI
Solaris	Solaris	Solaris
EVI-D70 (Sony)	Sony	EVI-D70
Receiver/MPU (Sungjin)	Sungjin	Receiver/MPU
ORX_1000 (SysMania)	SysMania	ORX_1000
C-CC501 (TOA)	TOA	C-CC501
DMP-1223 (Tokina)	Tokina	DMP-1223
KD6 (Ultrak)	Ultrak	KD6
KD6 Z-Series (Ultrak)	Ultrak	KD6 Z-Series
Vicon (Vicon)	Vicon	Vicon

78. Strong Password

All passwords must meet the following guidelines:
- Not the same as User ID
- Use 8 to 16 characters.
- Include at least three character types (uppercase, lowercase, numeral, special character)
(e.g jA38v2c4 or a1##sb32)

The Password CANNOT CONTAIN:
- number sequence (i.e 123, 321)
- alphabet sequence (i.e abc, cba, ABC, CBA)
- repeating characters (i.e 111, aaa, AAA)

79. TCP Event Type

Type	Description
alarm_in_on	This event type occurs when alarm in event start.
alarm_in_ignored_on	This event type occurs when alarm in event start for time of ignore interval.
alarm_in_off	This event type occurs when alarm in event stop.
motion_on	This event type occurs when motion detection event start.
motion_ignored_on	This event type occurs when motion detection event start for time of ignore interval.
motion_off	This event type occurs when motion detection event stop.
tripzone_on	This event type occurs when tripzone event start.
tripzone_ignored_on	This event type occurs when tripzone event start for time of ignore interval.
tripzone_off	This event type occurs when tripzone event stop.
tampering_on	This event type occurs when tampering event start
tampering_off	This event type occurs when tampering event stop
audio_on	This event type occurs when audio detection event start.
audio_ignored_on	This event type occurs when audio detection event start for time of ignore interval.
audio_off	This event type occurs when audio detection event stop.
pir_on	This event type occurs when pir dectection start.
pir_ignored_on	This event type occurs when pir dectection start for time of ignore interval.
pir_off	This event type occurs when pir dectection stop.
disk_on	This event type occurs when SDCard is inserted.
disk_off	This event type occurs when SDCard is ejected.
auto_tracking_on	This event type occurs when object dectection start.
auto_tracking_off	This event type occurs when object dectection stop.
loitering_on	This event type occurs when loitering dectection start.
loitering_ignored_on	This event type occurs when loitering dectection start for time of ignore interval.
loitering_off	This event type occurs when loitering dectection stop.
line_crossing_on	This event type occurs when line crossing dectection start.
line_crossing_ignored_on	This event type occurs when line crossing dectection start for time of ignore interval.
line_crossing_off	This event type occurs when crossing dectection stop.
object_detection_on	This event type occurs when object dectection start.
object_detection_ignored_on	This event type occurs when object dectection start for time of ignore interval.
object_detection_off	This event type occurs when object dectection stop.

※ “Ignore interval time” can be checked in the camera RemoteSetup and event web api cgi.